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Technology Transfer and Commercialisation for the European Green Deal

Final Report – January 2021

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1 DISCLAIMER: The views expressed in this report are solely those of the author(s) and do not necessarily represent the official views of the Joint Research Centre or the European Commission.

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Abstract

The rapid deployment of green technologies will be vital to put the EU on a path to meeting the European Green Deal objectives. Many green technologies with a great potential to contribute to the EU's green transition are currently being developed in research laboratories across the EU. How can we ensure that these novel technologies find the right conditions to mature and become widespread in the EU? What are the main barriers in the transfer of green technologies from the research organisations to the market? And what could facilitate the rapid commercialisation of green technologies?

This report on technology transfer and commercialisation for the European Green Deal sets out to answer these questions. A particular focus was placed on barriers and facilitators for green technologies in the areas of EU policy, intellectual property and financing. Four case studies were carried out in the fields of hydrogen, batteries, carbon capture and storage and artificial intelligence. The analysis resulted in policy recommendations on how to facilitate the technology transfer and commercialisation of green technologies. Giving green technologies the best chances to reach market maturity and have an impact in transforming our societies will take us a step closer towards achieving sustainability goals.

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EXECUTIVE SUMMARY

The rapid deployment of green technologies will be vital to putting the EU on the path to meeting its European Green Deal objectives. Many green technologies with a great potential to contribute to the EU's green transition are currently under development in research laboratories and universities across the EU.

This report analyses the process of technology transfer of green technologies from public research organisations to the market and identifies the main factors that may facilitate or hinder that process. The general findings about these factors are then confronted with the status in four case study areas on relevant green technologies: hydrogen, batteries, artificial intelligence and carbon capture and storage. Resulting from this analysis, the report draws a set of policy recommendations on how to facilitate the deployment of green innovation in the EU.

1 Overview on technology transfer and commercialisation

The report first sets the scene by introducing the basic processes of technology transfer and commercialisation (section 2). It then proceeds by giving a horizontal overview of relevant policy instruments (section 3.1) and the evolution of the EU's policy framework (section 3.2), followed by a closer look at the cornerstones of the technology transfer process.

Intellectual property ("IP"; section 4.1) is a major facilitator for the green technology transfer process insofar that it enables transferring the results of intellectual creativity from an inventor to the market. However, IP protection may also represent a barrier to the technology transfer process. Especially SMEs oftentimes lack the necessary knowledge of managing IP assets and do not have an IP strategy. Different national legal regimes and the territorial nature of intellectual property rights may complicate the IP management and make their commercial use challenging. The costs related to intellectual property protection remain high, while the future profitability of fundamental research is often uncertain, leaving many patents unexploited. Nevertheless, interviews conducted for the report revealed that stakeholders are less concerned with the challenges posed by IP than the literature would suggest.

Commercialisation is the next phase of the technology transfer process. After delivering technological innovation and ensuring its legal protection, commercialisation may be hindered by several barriers (section 4.3). Matching innovators, technology transfer offices and potential investors is less problematic than initially assumed, yet the successful conclusion of an investment decision between these parties faces several barriers. There is a noticeable information asymmetry between the parties and differing perceptions of the underlying risks. This is largely due to the perceived lack of clear-cut business cases as well as a high-risk exposure along both the value chain and the time horizon. Coordinated, blended financing is missing, which would be necessary to lower the risks when bringing innovation to market. Richer data for improved risk models is needed to facilitate private finance, while the management skills of SMEs and start-ups should be improved in order to improve financial planning.

With regard to **political and regulatory barriers** (section 4.4), regulatory uncertainty and lack of clear political commitment to green technologies can represent a substantial barrier to the commercialisation of new green technologies. Both investors and innovators need regulatory certainty and require coherent and coordinated policy choices. A fragmented regulatory framework and uncoordinated national approaches would risk hindering the exploitation of cross-border synergies in the EU.

2 Case studies on green technologies

The second part of the report examines technology transfer barriers and facilitators in four case study areas: (1) hydrogen technology, (2) battery technologies, (3) artificial intelligence, and (4) carbon capture storage and utilisation.

Hydrogen technologies (section 5.1), and in particular green hydrogen generated from renewable energy sources, have emerged as one of the key enabling technologies for the transition to a zero-emission society. As recent European and national hydrogen strategies highlight, there is a political impetus for further accelerating the deployment of hydrogen. EU-level initiatives, such as the Fuel Cells and Hydrogen Joint Undertaking or the newly launched European Clean Hydrogen Alliance, have been and will be crucial to boost the technology transfer and commercialisation of hydrogen technology. In this regard, European policy makers need to support the rapid scale-up of hydrogen production with a view to lowering the costs of green hydrogen and electrolysers to competitive levels. Further public action will be required to establish adequate infrastructures and to provide a fit-for-purpose regulatory environment. In addition, the public sector may need to help build up the end-user demand for green hydrogen and sustain the next generation of hydrogen innovation in its early stages.

As in the case of hydrogen, stakeholders in the **battery industry** (section 5.2) are cautiously optimistic about the future, as the EU has made a clear political commitment to bring the battery cell production to Europe. Significant results have already been achieved through intensive coordination between R&I and the industry (*e.g.* European Battery Alliance). This collaborative ecosystem enables investment and supports knowledge transfer. Policy makers should be careful, however, to keep the opportunities open to novel battery technologies as there is a clear risk of directing all attention to the current winning technologies (Li-ion batteries). Risks remain high for innovative start-ups in the field and public support is still needed to reduce the financial risk to investors and to foster integration along the whole value chain.

Overall, the battery industry may, however, serve as a good example for the deployment of other green technologies.

Artificial intelligence (section 5.3) is a general-purpose technology that has a strong potential to make a wide variety of processes or sectors more sustainable. On the opposite end, as AI is poised to become a mainstream technology, "greening" AI, such as developing more efficient AI algorithms, greening data, and greening AI hardware and infrastructure, will play an important role to limit environmental footprint of AI. The lack of a robust regulatory framework was brought up by stakeholders as posing a barrier to the development and commercial use of AI in the EU. Moreover, European stakeholder networks should be further facilitated to achieve stronger coordinated approaches to the use of AI and to incentivise greener AI innovation. Stakeholders highlight EU strategies such as the Digital Europe Programme, Horizon Europe, and regulatory sandboxes as potential facilitators for AI technology transfer and commercialisation.

The final case study on **carbon capture utilisation and storage** (section 5.4) shows that technology transfer in the field does not constitute a major challenge as industry players are the main driver of the innovation in this field. Commercialisation, however, faces numerous barriers. Public support is needed to significantly decrease the costs borne by first-movers and to de-risk investment and initial operation over the time horizon and across the value chain. Only a few demonstration projects exist and further processing or end-users are missing, which leaves investors without a clear business case. Public acceptance and political commitment must be strengthened further in order to create certainty for investors, whose return would materialise over a longer timeframe. The regulatory framework still needs to be improved to enable cross-border operation for the industry, which would benefit from operating as a part of a broader European network.

3 Recommendations

In its concluding section, the report sets out several policy recommendations (section 6.4), which aim at improving the transfer and the commercialisation framework for green technologies. The recommendations address key issues concerning the public policy framework for technology transfer of green technologies, the technology transfer process itself, and the financial environment and infrastructure. While these recommendations are general, successful technology transfer and commercialisation policies need to factor in the specificities of the technologies and industries concerned, as highlighted in the report's case studies.

The recommendations are summarised in the figure on the following page.

Public policy framework

01

Public policy should provide an enabling and stable regulatory framework ensuring long-term regulatory certainty and should be supported by a strong narrative and clear objectives (*e.g.* climate neutrality targets). EU-level regulation and coordination with Member States is needed to avoid regulatory fragmentation and ensure technical interoperability across borders, enabling high-impact green technologies to be scaled-up and deployed EU-wide.

02

The policy should follow a holistic, integrated approach along the entire value chain, from the early research stages up to mature technologies, reflecting the needs of green technologies with a systemic reach. R&I policy, industrial policy and environmental policy and their respective tools and objectives should not act in silos, but be joined up into an integrated public policy approach.

03

EU institutions should continue to play an active role in strengthening existing stakeholder networks and building up new ones, thereby bringing together industry, researchers, investors and policy makers (*e.g.* in industrial alliances and public-private partnerships).

Financial environment

04

Increase the availability of dedicated green funds, loan and guarantee schemes and promote public-private partnerships with a high public visibility to show a high level of political commitment and foster technology acceptance in investors and the wider public.

05

The EU should develop data reporting guidelines for green technologies and new risk assessment models that factor in non-financial risk, such as the effects of climate change. Financial institutions need to be given better toolkits to assess risks related to green investments.

06

Public support needs to bring down first movers' costs for green technologies. In this regard, public support should bridge gaps in private funding for high-impact technologies at low technology readiness levels. Financial support should cover demonstration projects and commercialisation (including CAPEX, OPEX and enabling infrastructure) through blended financing and guarantee instruments.

Technology transfer process

07

Increase the availability and use of real-world testing environments and regulatory sandboxes to permit new innovation to reach market readiness and ensure higher success rates when deployed to the market.

08

Open innovation models should be studied with a focus on the needs of various green technologies. In particular, it should be analysed whether pooling green patents would be desirable and would support the deployment of green innovation in certain sectors.

09

Technology transfer and commercialisation activities should form a prominent part of the design of public research policies, as well as one of the strategic priorities of research and technology organisations. To that effect, sufficient funding should be ensured both for the technology transfer process and for technology transfer offices.

1. INTRODUCTION

With the European Green Deal, the Commission outlined its plans to transform the European Union into a modern, resource-efficient and competitive economy where there are no net emissions of greenhouse gases by 2050. The unexpected outbreak of the COVID-19 pandemic has transformed our societies and our economy in an unprecedented way, but it has also created a window of opportunity to reshape our policies and accelerate the green transition of our economy as the EU recovers from the crisis. The new Multiannual Financial Framework (MFF) together with the recovery instrument, Next Generation EU (NGEU), mobilises EUR 1.8 billion in the next seven years with the focus on growth, resilience, research, digital and green transition. In the new MFF, 30% of the EU funds is allocated for fighting climate change, while significant share of NGEU, worth of EUR 750 billion, is planned to contribute to the goals of the Green Deal and to tackle climate change trough different funds, such as the Recovery and Resilience Facility. Therefore, there is a great financing potential foreseen for green technologies in the coming years, which makes it timely to further explore the transfer and commercialisation of these technologies.

The green transition of the EU economy in part requires a wider deployment of existing green technologies, such as wind energy, renewable energy grids, or biodegradable packaging for consumer products. However, at the same time there is also a need for new and improved green technologies to make the green transition possible. These green innovations can result either from the improvement of existing technologies or from the development of new technologies. Many technologies with a great potential for the realisation of the Green Deal are presently being developed in research laboratories across the EU. Giving these technologies the best chances to achieve market maturity and have an impact in transforming our societies is crucial for the EU's green transition and for maintaining a leading position in technological innovation.

The concepts of technology transfer and commercialisation describe the vital intermediating processes that facilitate the transition of new technologies from research laboratories to the market and the EU's wider society. To make technology transfer possible, a variety of important factors play a role. These factors include finance, know-how of the specific technology, awareness about intellectual property (IP) protection, and willingness of stakeholders to move the idea forward. The same applies to commercialisation, where decisions over IP rights, business models, markets, legislation, and sales come into play. These and more together can form an array of barriers and facilitating factors along the road of conceiving new technologies and their arrival and use in households, the public sector, and businesses.

The aim of this report is to investigate which barriers prevent green technologies from successfully going through the technology transfer and commercialisation processes; and which facilitators are enabling green technologies to succeed. In this regard, this report addresses key aspects of research and innovation policy, but it also touches upon linked areas of public policy, such as industrial policy.

The second chapter of this report briefly outlines the main parts of technology transfer and commercialisation, and the way in which the concept of "green technology" is approached. The subsequent two chapters present the overall policy landscape and the barriers and facilitators of technology transfer (in the areas of IP, legal matters, commercialisation of technologies, financial aspects, as well as political and regulatory aspects).

The fifth section of this report aims to identify barriers and facilitators pertaining to green technology through four case studies on: hydrogen technology, batteries, artificial intelligence and carbon capture and storage. The case study areas represent key technologies for the green transition and decarbonisation of the European economy, and they are closely linked to the vision of the European Green Deal:

"EU industry needs 'climate and resource frontrunners' to develop the first commercial applications of breakthrough technologies in key industrial sectors by 2030. Priority areas include clean hydrogen, fuel cells and other alternative fuels, energy storage, and carbon capture, storage and utilisation. "

"Digital technologies are a critical enabler for attaining the sustainability goals of the Green Deal in many different sectors. The Commission will explore measures to ensure that digital technologies such as artificial intelligence [...] can accelerate and maximise the impact of policies to deal with climate change and protect the environment ".[1]

The final chapter of the report presents conclusions and recommendations based on the findings from the case studies.

This report and in particular its case studies are based on extensive desk research and interviews with selected experts and stakeholders. Input for the horizontal part of the study was collected through desk research and semi-structured interviews with relevant stakeholders and experts. This included notably experts from the services of the European Commission, technology transfer, patent and intellectual property offices and organisations, university researchers and representatives of the financial sector.

In addition, the report is based on a survey on technology transfer and commercialisation of green technologies, conducted among the members of the European Technology Transfer Offices Circle[2]. Furthermore, stakeholders in the selected case study areas were identified with the help of experts from the European Commission. Additional interviewees were identified applying the snowball sampling approach. These stakeholders and experts were consulted in semi-structured interviews focusing on their respective case study area. In general, interviews were conducted under the condition of anonymity.

This report is not meant as a conclusive state-ment but as a contribution to current debates on the deployment of green technologies in Europe.

2. TECHNOLOGY TRANSFER & GREEN TECHNOLOGIES

Technology transfer is a vital part of technological innovation. Technology transfer refers to the process where results and ideas (and associated skills and procedures) that originate from scientific and technological research are transferred to the marketplace and the wider society.[3]

The technology transfer process, that starts at universities and research centres and ends with societal use, is composed of a complex value chain which includes:

- **Discovery** of new technologies at research institutions;
- **Disclosure** and **evaluation** of new technologies;
- **Intellectual property protection**;
- **Product marketing** and **licensing**;
- **Product development** and **deployment**;
- **Public use, sales and financial returns**.

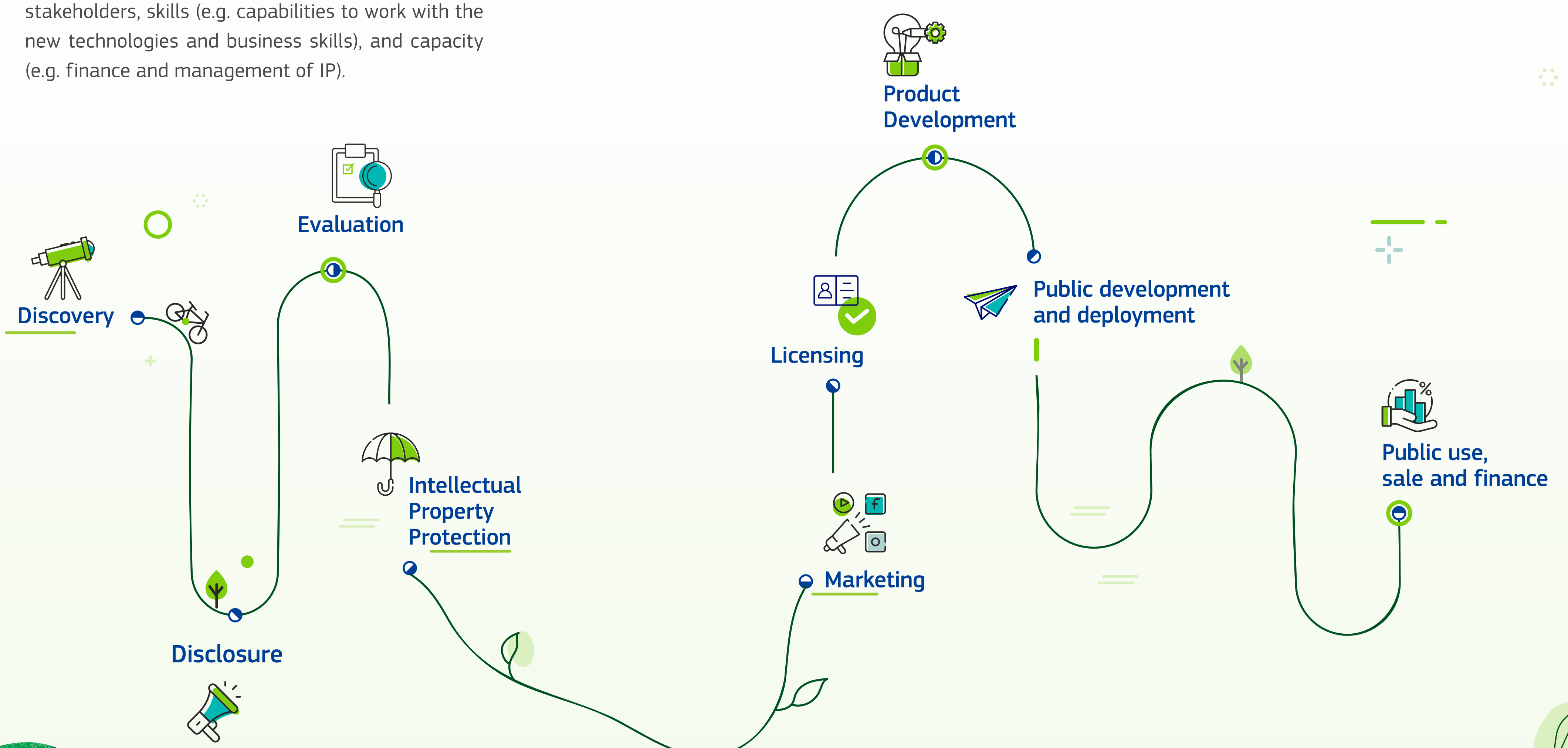
Technology transfer processes involve a plenitude of stakeholders beyond the teams that are executing R&D&I activities. For technology transfer to succeed, there is a need for awareness and willingness from stakeholders, skills (e.g. capabilities to work with the new technologies and business skills), and capacity (e.g. finance and management of IP).

In this report, technology transfer will be delineated to the (often seemingly linear) process that is described above. The report will not refer to technology transfer as used in the international trade arena, which is the process where technology is transferred between geographical areas; often from a developed to a developing state.

It is also delineated as a subset of knowledge transfer, a broader term that oftentimes refers to "the flow of virtually all knowledge in an organisation and the attempts to create, capture and manage that knowledge".

The definition of "green technologies" has a range of conceptions in the academic literature. In this report, green technologies will refer to all the technological innovations that have a potential to substantially contribute to the aims of the European Green Deal. In general, these are the technologies that can support the transition to a more sustainable path in the EU and to reach the objective of climate neutrality by 2050.

Figure 2: The value chain of the technology transfer, which links research to its eventual societal deployment. Source: European Commission[4].



3. TECHNOLOGY TRANSFER AND COMMERCIALISATION POLICY

What instruments are available for governments to support technology transfer and commercialisation? Additionally, what was the policy response at EU level thus far? To answer these questions and to provide a backdrop for this report, the following section gives a brief overview of relevant policy instruments and the recent EU policy on the matter.

3.1 Overview of policy instruments

In the context of technology transfer, policy can be considered as the sum of all public policy instruments deployed at various levels of government to support technology transfer. Policy instruments include a wide range of financial support instruments (*e.g.* for academic spin-offs), regulatory instruments (such as the design of the IP rights system or open data provisions), and soft instruments (such as awareness raising campaigns or guidelines). In this regard, financial instruments may also address issues related to the commercialisation of technologies.

A recent OECD report has identified 21 types of policy instruments (including the aforementioned examples) that can support science-industry knowledge transfer. Given the close link, this classification can be applied just the same to technology transfer.[5]

The OECD report classifies the identified instruments along four dimensions:

1. whether they are financial, regulatory or soft instruments (see examples provided above);
2. whether they target primarily firms, research and technology organisations, individual researchers or research groups;
3. the type of knowledge transfer channels that are being addressed;
4. the supply- or demand-side orientation of policy instruments.

Most countries adopt similar instruments but with noticeable differences regarding the prioritisation of measures, as well as the detailed design of the instruments.[6]

In this context, policy makers also need to consider carefully interaction effects among policies. Unnecessary complexity and potential inconsistencies in policy instruments need to be avoided. A well-designed policy framework should facilitate all major steps of technology transfer and commercialisation, follow a step-by-step approach when the sequence of introduced policy instrument matters (pre-conditionality) and capitalise on potential synergies between policy instruments.

The needs as well as the effect of applied instruments may vary from sector to sector. For example, the overall situation may be quite different in green technologies compared to information technology (*e.g.* hydrogen technology compared to quantum computing).

Moreover, one must assume a substantial interaction effect with the policy mix in place.[7] Relevant elements include the macroeconomic context, the characteristics of the business community (both at country and at sectoral level), the characteristics of universities and research institutes as well as social and cultural values (*e.g.* related to entrepreneurship).

This means that the historical evolution of research and technology organisations and the private sector may influence the current performance of the technology transfer environment.

Altogether, a well-designed policy framework is a key factor to enable technology transfer. In the European context, this requires an efficient balance of policy instruments, at the EU-level as well as the national and regional level.

3.2 Recent evolution of EU policy

In the EU, research and technological development is a shared competence between the Union and its Member States. Article 4(3) of the Treaty on the Functioning of the European Union states:

"In the areas of research, technological development and space, the Union shall have competence to carry out activities, in particular to define and implement programmes; however, the exercise of that competence shall not result in Member States being prevented from exercising theirs."

The Commission's competence is thus somehow restricted, as its actions on technology transfer have to go hand in hand with the Member States' own interventions.

Building on subsequent Communications[8], the European Commission has developed its policy on knowledge/technology transfer since the development of the European Research Area (ERA) in the 2000s.[9] In particular, the Commission has used the following instruments:

- setting-up expert groups on knowledge/technology transfer and related matters to support the work of the Commission and develop *e.g.* guidelines and recommendations;
- developing European networks, such as the European Technology Transfer Offices (TTO) Circle, a network of the technology transfer offices of Europe's largest public research organisations;
- providing funding to improve the technology transfer capacities of Member States;
- including aspects of technology transfer in Horizon 2020 and the EU Research Framework Programmes;
- development of technology transfer capacities and expertise inside the Commission, *e.g.* with the build-up of the Competence Centre on Technology Transfer;
- mainstreaming of technology transfer and awareness raising (*e.g.* via Commission Communications and recommendations).

In 2007, the Commission adopted a Communication on improving knowledge transfer between research institutions and industry across Europe.[10] The Communication was followed-up in 2008 with a Recommendation on the management of intellectual property in knowledge transfer activities and a Code of Practice for universities and other public research organisations.[11] Subsequently, the 2010 Communication on the Innovation Union featured a chapter on "getting good ideas to market" committing to various financial support actions.[12]

In the context of the development of the European Research Area (ERA), technology transfer was covered in ERA's priority five on the

"optimal circulation, access to and transfer of scientific knowledge, including 'knowledge circulation' and 'open access'".

2012 marked the Communication on a Reinforced ERA Partnership for Excellence and Growth, aiming at completing ERA by 2014. While progress on ERA was achieved, the efforts continued in the context of the European Research Area Roadmap 2015-2020.

In 2019, the ERA Progress Report noted that while the overall progress on the ERA roadmap had slowed down, indicators for priority five indicated a stable improvement. At the same time, the report concluded that

"the transfer of research results to the market and closer collaboration between industry and academia still need to be promoted in many ERA countries"[13].

In 2020, a new Commission Communication on the Future of Research and Innovation and the European Research Area was announced, reflecting the call of the Council of the European Union for a new Communication on the future of ERA. The Communication on "A new ERA for Research and Innovation" aims to provide a forward-looking orientation for ERA, 20 years after its establishment.[14]

Noting that the progress regarding ERA has slowed down, the new strategy tries to give a new impetus to the project, not the least to support the Green Deal and the digital transition. In this regard, the ambition is to put the European research and innovation policy on track to play its role in overcoming challenges related to societal, ecological and economic issues as well as the impact of the COVID-19 pandemic. For this purpose, the Communication sets four strategic objectives:

1. Prioritising investments and reforms to accelerate the green and digital transformation and to increase competitiveness as well as the speed and depth of the recovery;
2. Improving access to excellence to enable stronger R&I systems across the whole of the EU where best practice is disseminated faster across Europe;
3. Translating R&I results into the economy to ensure the EU's competitive leadership in the global race for technology while improving the environment for business R&I investment, deployment of new technologies and enhancing the take up and visibility of research results;
4. Deepening ERA to further progress on the free circulation of knowledge in an upgraded, efficient and effective R&I system.

Progress along these objectives should be achieved via a new ERA Roadmap, accompanying the revised ERA Strategy. In particular, the third objective of the Strategy on translating R&I results into the economy has a strong link to this report and can be interpreted as a push to enable the technology transfer and commercialisation of key novel technologies. To this end, the strategy stresses the concept of knowledge valorisation, emphasising thereby a renewed focus on the transformation of intellectual property and research results into market ready solutions creating societal and economic value.[15]

In this regard, the new ERA Strategy announced that the Commission would: [16]

- Develop and test a networking framework in support of Europe's R&I ecosystems to strengthen excellence and maximise the value of knowledge creation, circulation and use by 2022;
- Develop guiding principles for knowledge valorisation and a code of practice for the smart use of intellectual property by the end of 2022 to ensure access to effective and affordable intellectual property protection.

4. TECHNOLOGY TRANSFER FACILITATORS AND BARRIERS

Many factors determine the chances for successful technology transfer and commercialisation. We can think of these factors both as facilitators and as barriers to the process. Policy instruments aim at shaping these factors in a positive way to facilitate the uptake of innovative technologies. This section describes how key factors - such as finance or intellectual property - can influence technology transfer and commercialisation.

4.1 Intellectual property

Potential Barrier to Technology Transfer Intellectual Property

- Besides acquiring patent rights, additional knowledge (such as technical know-how and trade secrets) is often indispensable to reproduce and commercialise a green invention;
- Substitutability of green technologies leads to a fiercer competition among technological solutions, resulting in lower impact of each patent and uncertain return on investment;
- Patenting green technologies is often connected to comparatively greater investment risks, as a high percentage of green patents end up commercially unexploited;
- IP protection is costly, which is further exacerbated by the territorial nature of IP protection and the resulting need to independently maintain and enforce patents in each jurisdiction;
- A lack of adequate knowledge and IP management capacity persists on the side of some research technology organisations and SMEs. Among others, this results in difficulties with developing an adequate IP strategy;
- The inaccuracy and inherent complexities of IP valuation remain a major obstacle to the emergence of IP as a tradable asset class;
- Improvements are needed to the engagement of both research and industry representatives in collaborative R&I and open innovation platforms;
- The existent green tech IP collaboration platforms have often proved suboptimal and have not met their objectives of supporting wide dissemination of green technologies.

Despite certain aspects of IP framework which may negatively impact the efficiency of the technology transfer process or may act as barrier to the process (see the box above), in general, IP protection is the first major facilitator of technology transfer (or in fact its prerequisite). Once the ownership of intellectual property is ensured, its owner has a monopoly over the innovation and can sell it or license it to any party. Therefore, IP protection plays a key role – both for innovators to recoup the investment in R&D, and for businesses involved in commercialisation because it enables them to reap the benefits of the innovation to the exclusion of other competitors.

Regarding technology transfer in general and green technology transfer in particular, the most prominent IP rights are patents and trade secrets. Other IP rights, such as copyright for codes and computer programmes, plant breeder's rights, utility models, or database rights, play a role in specific instances.

Patents

Patents are the main instrument used by European universities and public research organisations for exploiting their inventions commercially. According to a study by the European Patent Office ("EPO"), European universities and public research organisations already exploit more than a third (36%) of their inventions for which they have filed a patent application with the EPO, with another 42% of their inventions planned to be exploited.[17]

Patent protection only emerges after a patent is registered and granted by a relevant patent office in every patent jurisdiction where protection is sought. This means that, despite some international efforts at harmonisation,[18] IP rights to a single intellectual product may have a different scope of protection in each country and be subject to different registration and enforcement procedures. Such legal fragmentation frequently results in additional expenses in maintaining and commercialising IP assets (multiple registration and maintenance costs, costs of legal advice, patent counselling costs, costs linked to the enforcement of one's IP rights etc.).

The legal fragmentation of relevant IP laws demonstrates itself also in the default IP ownership regime for university research. In the absence of any agreement between university and its researchers, most Member States attribute the ownership of inventions developed at universities to the university. However, there are marked differences between the national regimes. Moreover, in some jurisdictions[19], it is not the university but the inventors who shall by default own rights to their research results (the so-called "professor´s privilege").

Trade secrets

In technology transfer, patent-related rights without an access to associated or collateral know-how are often insufficient for successfully commercialising technology.

While patents tend to have a narrower scope to demonstrate the inventiveness of the innovation, trade secrets cover more comprehensive knowledge about an invention than carefully drafted patents. Therefore, trade secrets can also act as an enabler of technology transfer, usually alongside patents.

Apart from their role in providing complementary knowhow to patents, trade secrets also play a role in protecting the invention before patent filing. Businesses, especially SMEs,[20] frequently rely on trade secrets instead of patent protection, because trade secrets have no registration and maintenance cost, and are effective immediately and for unlimited time. However, it needs to be highlighted that trade secrets are protected only to the extent that they are kept secret. Trade secrets do not prevent competitors from engaging in reverse engineering and parallel innovation. To protect the confidenti-ality of information, any technology transfer thus normally involves entering contractual non-disclo-sure obligations (NDAs).

IP in technology transfer

Research technology organizations can exploit their IP rights in several ways. The approach depends on factors such as the industrial applicability of research, the ability of the invention to generate income or the capacity of the inventors to commercialise it. According to EPO data, licensing is the most common channel of commercialisation of inventions developed at the European universities and public research organisations (70% of the commercialised university patents).[21]

Besides licensing, another technology transfer route is to sell the given IP right (accounting for 9% of the commercialised university patents[22]). Notably, by selling - assigning - the IP asset to another entity, the inventors typically lose their right to exploit it and such right is vested in the new owner. However, IP rights can also be transferred as a contribution in kind to a university spin-off where the inventor may be actively involved.

Moreover, it becomes increasingly common to use various join exploitation schemes and R&D co-operation models (14% of the commercialised university patents).[23]

IP in green technology transfer

Even though green technologies may differ from one another, certain common features can be identified.

As compared *e.g.* to patents in chemical substances, where a disclosure of chemical composition may be sufficient to reproduce innovation, green technology is typically difficult to reproduce only on the basis of a patent without the access to the relevant knowhow or trade secrets.

While there exists small and less-capital intensive green technologies, a good number of green technologies is a result of larger capital-intensive engineering projects. These green technologies, especially renewable energy technologies, are typically characterised by a higher degree of complexity than their non-green alternatives or than other technologies.

As a result, they usually consist of several off-patent and patented solutions to which IP rights can be held by several different owners. This may compli-cate their diffusion - a challenge that applies especially in the case of the incipient green technologies, such as those for climate change adaptation or carbon capture and storage.[24]

Moreover, one technical problem often has more than one technical solution. This encourages competition among green patents that often leads to more reasonable royalty rates and incentives to cooperate.

As a result, there is no systematic evidence at hand that IP would restrict the access to green technologies as may happen in other patent-dense sectors (e.g. pharmaceuticals or nanotechnologies), where the undue width, overlaps or general concentration of patents may cause “patent thickets” which can complicate the patenting and commercialisation of subsequent innovations by creating an impenetrable thicket of exclusionary rights.

Overall, green patents represent a limited share (8.6%) of all patents.[25] A patenting boom in green technologies happened in the period between 2005 and 2013 when patent families in green technologies almost doubled worldwide (especially in solar energy, fuel cells, wind energy and geothermal energy). The growth in patenting activity has slowed down since, mainly due to a sharp decline in patenting in the solar energy sector that had matured during that time.[26] Despite this general slowdown, certain technologies of increasing importance for the clean energy transition have maintained or even increased levels of patenting activity.[27] Innovation in greener transportation, for example, continues to grow steadily.

The same patentability criteria apply to green technologies as to other technologies.[28] However, certain countries differentiate in favour of green patent applications by granting them a more favourable procedural regime (see the table Fast-track green patent applications).

Fast-track green patent applications

Certain countries differentiate in favour of green patents by granting them a more favourable procedural regime. Examples of such regimes could be reduced patent filing fees or accelerated patent registration procedures.

The United Kingdom introduced an accelerated patent registration model “Green Channel” for all environmentally friendly inventions in 2009. Similar programs also exist in Australia and Canada. Brazil, China, Israel, Japan, and South Korea have fast-tracked patent registration procedures for some green technologies.

These fast-track procedures may shorten the patent registration procedure by several months or years.

Despite this, the appeal of the fast-track procedures is inconclusive with anything between 0.75% and 20.9% of eligible patents requested under these fast-track regimes (depending on a country). The standard patent examination period might well prove to be advantageous to inventors as it gives them more time to adjust patent claims. It also delays public disclosure of a patent, as well as the payment of patent maintenance fees.

On the other hand, early patents may give a competitive advantage to inventors, especially SMEs, allowing them to have an early start in searching for capital or commercial partnerships. This may prove to be useful especially in sectors with highly dynamic patenting activity, such as e-mobility.

More information at: https://www.wipo.int/wipo_magazine/en/2013/03/article_0002.html[29]

As opposed to (for example) pharmaceutical inventions that often hold patents in most jurisdictions worldwide, green technology is typically patented internationally only in a handful of major patent jurisdictions (such as the United States, Europe, Japan, South Korea, and China).

European innovators are frontrunners in green technologies.[30] In the past two decades, the patenting rates for clean energy technologies have increased roughly by 20% per year, ahead of traditional energy sectors.[31] Notably, Europe´s innovators have strong IP portfolios in technologies such as carbon capture and storage, water and waste treatment and climate change adaptation.[32] European innovators also hold substantial shares of green patents in transportation, buildings and smart grids.[33]

The European Patent Office has introduced a patent classification scheme specifically covering climate change mitigation and adaptation against climate change technologies (“Y02” classification) and smart-grid technologies (“Y04” classification). This classification, including its subcategories, plays an instrumental role in easing access to patent information about green technology.

While Europe scores highly in green tech research results, research excellence does not smoothly translate into commercialisation excellence. Indeed, according to OECD data, Europe develops 28% of world´s green innovation while only 15.4% of that green technology ends up patented.[34] This illustrates the European “research paradox” – where relatively high R&D intensity in green innovation sector does not translate into successful commercialisation and industrial performance.

Green technology and open innovation

Since succeeding in isolation is rare, the development and diffusion of complex green technologies often require collaboration of various actors in the given field.[35] The environment of collaboration and sharing knowledge facilitates the creation of innovation, adaptation of existent innovation to local conditions or wider diffusion of innovation.

The technology transfer and commercialisation actors thus oftentimes enter in a complex system of mutual relations where innovation is co-created, rather than produced by one and licensed to another party.[36] Innovation can thus originate from joined activity of multiple actors, who cooperate *e.g.* under a collaboration agreement, joint research agreement, research collaboration agreement, or under the umbrella of a joint venture.

In such arrangements, it is crucial that the collaborating parties clarify legal title to future intellectual assets, division of revenues and other rights and obligations that will emerge from the cooperation.

Certain businesses prefer to be involved in university research from early on, rather than only engage later when the technology is ripe to be transferred or licensed. Through such cooperation, public research can be more focused and address specific market needs for innovative green solutions. Businesses get first-hand access to research results, as well as to concepts and practical and theoretical knowhow behind the invention which are often indispensable to commercialise complex green technologies.

To further facilitate the exchange among inventors, several industry-specific platforms have been created gathering both research bodies and public and private entities (*e.g.* European Battery Alliance or prospectively Clean Hydrogen Alliance).

Another example of open innovation is the act of simply sharing the invention with others by placing it in public domain. This can be done by either not patenting it altogether (or discontinuing with the maintenance of a patent) or by a non-assertion pledge. Such a pledge (not to assert rights against someone who uses their patent in compliance with given conditions) materialised *e.g.* in the sector of electric cars (by Tesla and Toyota) or in biotechnology. From a commercial perspective, by letting others reproduce the technology, this technology may get a new market boost and its general uptake may make it a standard in the given field.

There are also initiatives aiming at creating marketplaces for green patents or IP exchange platforms to share green technology and thereby improve its dissemination. One such green patent marketplace is WIPO GREEN that was set up by the World Intellectual Property Organisation in 2013. The Danish Patent and Trademark Office also runs an IP marketplace.[37]

WIPO Green

WIPO GREEN[38] is an online platform for technology exchange established in 2013 as a public private partnership to promote the diffusion of green technologies through gathering the information on global technology results and needs for green solutions. WIPO GREEN´s database thus helps to match the supply of green solutions with demand, thereby bridging the information gap between research and industry. The technology transfer mode is left to the discretion of parties. Currently, there are 3800+ listed technologies and needs in the WIPO GREEN database and it has 1500+ users around the world.

Registering a technology or need on the platform is free of charge and WIPO GREEN also provides additional support for technology transfer — *e.g.* by maintaining a database of experts who offer pro bono services, publishing guides or offering acceleration projects.

The platform can however only be as useful as useful is the input therein. As is the case with green technology in general, the patent information is often not enough to reproduce the technology – non-patented knowhow is often needed, as well as understanding of the practical applicability of the patents.

Indeed, it still remains a challenge to convey all the relevant information comprehensibly for a potential investor to assess whether a certain invention would be useful in their context.

While the green technology marketplaces for green solutions highlight the existing green solutions and gather information about market needs, other forms of IP collaboration can go further, actively organising patents and bundling them. As a result, all the patents needed for one technology could be licensed at once and transaction costs could be limited. One such solution is patent pools.

Green patent pool

A patent pool is an arrangement in which patent holders bundle their patents and license them 'en bloc' to one another or to third parties. The possible benefits could entail cost and process efficiencies, lowering transaction expenses, tackling the issue of patent thickets and royalty stacking, establishing a standard in the field, or reducing potential legal issues and likelihood of litigation.

Patent pools were successfully used in the past in different contexts – especially in standardized sectors with an abundance of SEPs. Instead of pooling patents essential to a standard, the patent pool could facilitate the access to the technology, simplify out-licensing and boost cooperation among patent holders. Therefore, rather than a SEP patent pool, a more pertinent model for a green patent pool would be the Medicines Patent Pool39 – an organization facilitating the diffusion of essential medicines in the developing world.

Based on this model, it could be further analyzed whether there would be a demand for a "green patent pool" in certain specific areas of green technology.

The scope of such a patent pool may focus on a specific outcome (such as wind turbines, charging of electric vehicles) or on a certain area (such as photovoltaics).

While exploring the desirability of patent pools in certain green sectors could be worth a detailed analysis, there are several practical constraints that need to be considered in case any platforms for green tech, patent pool or marketplace are set up. In particular, lessons should be learned from the earlier unsuccessful attempts at pooling green technologies – Eco-commons and GreenXchange.

Green Xchange

The GreenXchange was an online open innovation platform launched in 2010 by Nike and Yahoo! along with eight other organizations, where the participants agreed to share their IP in order to promote open innovation for sustainability. They could choose one of the several licensing models based on Creative Commons, including a royalty-free license.

The GreenXchange gathered over 400 patents. While four patents were pledged by universities, the platform had only limited success in reaching out to public research institutes. It turned out that to be of true value for universities, the platform would have to maintain a large portfolio of patents. Moreover, the project showed that both businesses and universities were more interested in gaining access to the knowledge and people behind the creation of the patents than in simply obtaining the patents without the knowhow.

The organizers also encountered several other problems in realizing the objectives of the platform. After its initial launch, it gathered only a limited number of additional patents and became heavily attached to Nike, its founder and main patent contributor. Moreover, it turned out that most of the participating companies would license their patents only under the condition that the patents would not be used in commercial products. Finally, the project lacked resources to maintain the platform and significantly reconceptualize the GreenXchange model to be more focused on relation-building around assets than on the patents.

The project was discontinued in 2011.

For more information on GreenXchange, see Open innovation for sustainability: Lessons from the GreenXchange experience.[40]

Eco–patent commons

Another initiative for sharing patents, Eco-Commons, encountered similar problems like GreenXchange.

Eco-Commons was an initiative of several multinational corporations to share with each other green patents that were otherwise lying fallow. The eco-patent commons managed to gather over 200 patents in the pool. The participants could access the pooled patents freely for the exchange of the same commitment on their part. Only patents with "environmental benefits" of certain IPC categories could have been placed in the commons.

However, the project was unsuccessful and was closed in 2016 after eight years of its existence. The main problem encountered was the limited usefulness of the patents pledged. These patents had generally a low value and were not chosen to address a particular need or demand of the others. The companies' motivation to keep maintaining the patents pledged to the pool as an exchange for getting the access therein was thus negatively impacted. Even though there were no statistics about the usage of patents from patent commons, it is safe to assume that the pool has not reached its goal of improving the uptake of green solutions. These problems, combined with the mismanagement and limited usefulness of patents without the related knowhow and personal interaction, finally led to the closure of the program.

For more information see Sharing Technology to Meet a Common Challenge.[41]

The design of any green patent pooling or patent collaboration platform would have to factor in the earlier experience with such endeavours.[42]

In particular, due to the complexity of green technologies and frequent substitutability of green technology patents, patents play a smaller role in green technology than in other sectors. Most platforms for patent management and exchange thus run short of the main facilitator of the diffusion of green technologies – the access to the persons behind the invention and to their knowhow.

Any platform aiming at the diffusion of green technologies would thus have to include not only patents but also additional know-how and copyrighted materials (manuals, software etc.) needed to replicate the invention as well as opportunities for interpersonal interaction to boost the informal knowledge spill over. This could involve networking, conferences or various pitching events.[43]

Rather than a general marketplace for all green technologies, it would be better to focus on technologies specific to each sector.[44] The platform would have to be user-friendly and make the matching between research supply and market demand easy, *e.g.* by including practical examples how a particular patent could be used, by functionally grouping the patents, including information about the licensing model envisaged by the patent holder and about the maturity of the technology. The platform could also include advice on the legal regime in various countries for commercialisation of the technology, national restrictions, and requirements as well as references to funding opportunities. This platform could be linked with the Horizon Results Platform and European Enterprise Network to ensure link with the research community, as well as larger companies and SMEs.

In conclusion, all the above-mentioned examples document the progressive abandonment of the *'linear model'* of technology transfer in favour of greater research collaboration between university research and industry. As will be illustrated in the case studies, these university-industry linkages significantly facilitate technology transfer of new green technologies to the market.

4.2 Legal aspects of the technology transfer

Potential barriers to technology transfer: Legal aspects

- Complexity of the existing legal rules (requiring expertise in several fields, including legal and technical expertise), costs of external counselling.

Besides the registration of and disposition with IP rights, the TTOs, investors and entrepreneurs in green technology have to grapple with several other legal aspects affecting technology transfer. These aspects include legal regimes governing the establishment, incorporation and modification of business corporations, environment-related tax regimes and tax incentives, business licences, competition law, contract and licensing law, (product) liability law and insurance, necessary authorizations for the product placement, and safety and documentation requirements. Each area of green technology is also affected by subject-specific rules (*e.g.* emission trading systems, access and property rights to data and privacy in case of AI, requirements for the selecting sites for CCS etc.).

Even though these legal aspects may play a role in technology transfer as both facilitators and barriers of technology transfer (and may influence the value of the technology transferred), a close analysis of these aspects would go beyond the scope of this report.

Since ensuring the compliance with all the legal requirements in place may be a hurdle to trial a new-to-market technology, regulatory sandboxes were introduced. Regulatory sandboxes, *i.e.* derogations to otherwise applicable regulations and requirements, help innovators to trial or bring to market innovative technologies, products, services, business models or procedures without having to comply with certain rules. Regulatory sandboxes have already been used in energy sector. They are nevertheless not unique to this sector – they have also found their use in supporting the development of novel techniques of waste recycling (derogations from otherwise applicable rules on waste disposal) and can be a useful tool in trialling other market-ready innovations.

With regard to standardisation, standards in the field of green technologies may have a dual impact as both a barrier and facilitator. While standard setting may set in stone standards and practices desirable for safety, sustainability and interoperability of technologies and may boost innovation to meet such standards[45], it may also stifle research on the alternatives that do not follow the given standard. Therefore, it should be done with care, especially in green technology sectors that are not mature yet.

4.3 Commercialisation of technologies

Potential barriers to technology transfer: Financial aspects

- Information asymmetry between innovators, TTOs and investors;
- Diversity of evaluation methods and lack of transparency on pricing;
- Often IP is seen as a tool for legal protection instead of a commercially exchangeable asset. Diverse methods of IP valuation;
- Lack of early-stage finance of technologies and drying up of public resources before concept development;
- Lack of coordinated financing and de-risking across the value chain and investment time horizon;
- High initial costs for first movers and lack of business case;
- High dependence on policy makers, but insufficient political commitment to certain technologies and policies;
- Little available commercial loans by banks and no public guarantor facilitating such lending activity;
- Lack of business knowledge and management skills in universities' / innovators' spin-off companies. Immaturity of start-ups and unclear financial structure;
- Lack of available historic data for improved risk assessment models;
- Public support to industry for developing and deploying green innovation may be limited by competition rules (especially mature technology and larger businesses).

Financing oftentimes plays a key role in allowing new innovations to successfully go through the technology transfer and commercialisation process. The need for financing generally starts at the early stages of innovation, where researchers at universities must have access to proper financing instruments in order to fund their research project. At the early stage of the process, research and innovation projects are usually funded by public research funds such as the Horizon 2020 programme (or philanthropic programmes).

Even though access to financing during the initial research stage is a challenge in itself, funding of research is beyond the scope of this report, which focuses on the technology transfer and commercialisation stages of green technology development. Hence, when we talk about financing technology transfer and commercialisation, our scope excludes the early-stage financing of research projects and we put access to finance in the centre of our discussion from the point when a concept of the technology is available.

In light of the aforementioned, three stages of the technology transfer process can be identified when access to finance becomes challenging:

- Valley of death and proof of concept;
- Licensing or exchanging of technology patents and IP rights;
- Upscaling technologies to commercially viable business products.

Valley of death

The *"valley of death"* is a gap in access to finance often referred to in the literature as a barrier to technology transfer and commercialisation[46]. In general, there are numerous public sources of finance available for research projects up until the point where the project reaches the publication of innovation. But the development of a working concept and then the creation of a viable product are not part of this type of funding. In theory, once innovation is patented, there are investors on the market searching for such IP assets with great potential that can be commercialised with return in the future. However, most often they are willing to invest in these technologies only when they have reached a certain level of maturity, *i.e.* when there is a well-developed prototype. This is due to the numerous IP-related financial risks that will be discussed below. As a result, a finance gap occurs in-between the stage of patent creation and the stage of having a well-developed prototype. This gap commonly poses a barrier for the transfer of innovative technologies since in lack of sufficient financial resources the development cannot continue. To address this, there are a few European solutions (*e.g.* EIF loans), that facilitate projects by providing special finance to projects that have an initial concept but that are not yet a developed commercial prototype.

Furthermore, the literature suggests[47] that the valley of death is less of an issue in the United States, because US-universities develop their innovations much further before looking for ways to commercialise them. Here the question arises, whether this is due to the particularities of the education systems or to the design of public research programmes.

Commercial use of IP assets

The second challenging point for access to finance comes during the technology transfer stage where the innovation has been patented and there is a working concept that can be presented to investors. TTOs and patent owners have to decide what would be the suitable tool to create revenue: should the patent be licensed, sold as a whole or a spin-off be created? This requires an understanding of the value of the patent and what revenues can be expected from it in the future. The exact value of a patent is hard to define. Since usually an innovation is new and unique, there is no market, competitor or substitute products or technology, against which the market value of the patent could be compared.

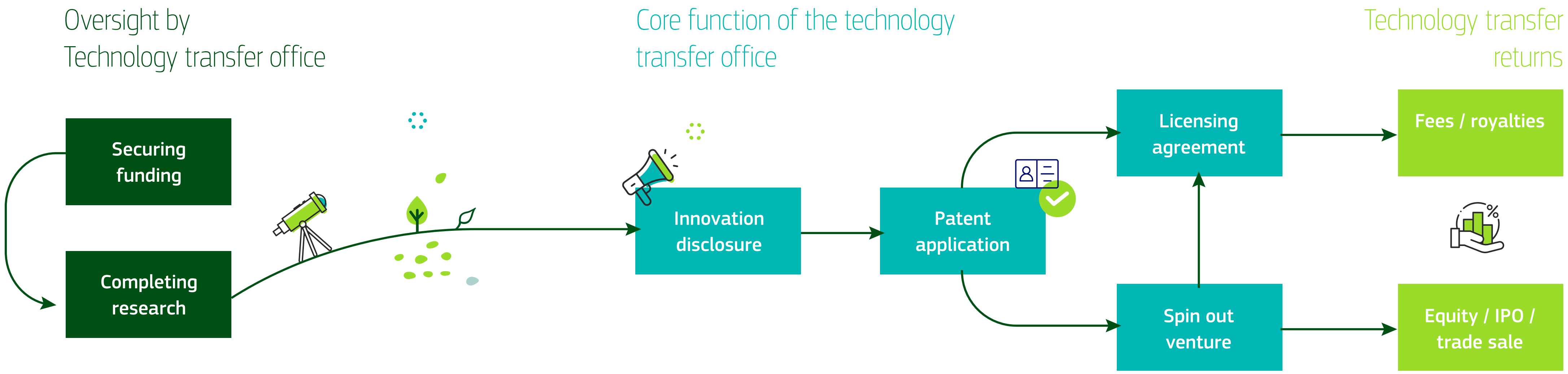


Figure 4: Commercialisation of technologies.
Source: Seget, S. (2008)

There are different methodologies to estimate the value of a patent. TTOs can apply:

- Cost-based approach – which values the patent based on the cost of its creation but neglects future expected revenues;
- Market-based approach – comparing IP assets to the price of similar assets licensed and sold on the market;
- Income-based approach – a discount rate incorporating potential risks in order to estimate the future generated revenue from the IP asset;
- Qualitative approach – subjective valuation by inventors factoring in different strengths and weaknesses of the innovation with weighting and scoring schemes[48][49].

The differences between these approaches, their individual weaknesses and the information asymmetry hinders TTOs and investors from successfully matching investments with the right IP assets. Often innovators struggle to find the right investors, since investors view their inventions as too risky opportunities, which are overpriced, and the attached future revenue streams are uncertain or difficult to estimate.

One of the main barriers remains the lack of access to private equity once the innovation has been patented, which is due to the underdeveloped nature of the IP market. Although there is a general trend showing that companies' assets are increasingly intangible, these assets – including IP rights and patents in particular – remain highly illiquid on the market. Investing in IP and technological innovation is considered to be risky by investors, who struggle to measure and price the attached risks accurately. Therefore, they will be less willing to invest in these assets or there will be a price mismatch between investors and the IP assets priced with the aforementioned methods. There is a need for increased transparency in order to have a better understanding of the attached risks by all players on the market.

Commercialisation of green technologies

Interestingly, the consultations with financial stakeholders and industry representatives showed that IP valuation and its commercial use is not as much of an outstanding barrier as literature suggested. Perhaps there is room for improvement in terms of better understanding the commercial potential of IP assets rather than using it as a legal instrument and making it more liquid and transferable.

Nevertheless, from the investors' perspective the biggest challenge is to be able to assess and estimate the risks associated with green technologies based on the available scarce present and historic data. Apart from the risks related to the technical feasibility of technology, it is first and foremost due to the lack of existing business case.[50] As these technologies are being newly deployed, investors see a high degree of uncertainty around them. It is often difficult to understand the viability of revenue streams, while the stability of sufficient supply and uptake of the final product on the demand side of the market remains foggy. On the one hand, there is a substantial risk regarding the time-horizon of the investment. Investing in green technologies is considered as a longer-term investment with higher dependence on public policy than other more conventional technologies, which carries higher risk potential. Investors need a stable regulatory environment with foreseeable policy strategy. There are numerous emerging green solutions under development or in the demonstration phase, and they often compete not just against older polluting technologies but also against each other. These parallel development processes are understandable from the policy makers' perspective, since they cannot bet everything on a handful of potential green technologies. However, investors would need assurances that the policy makers and the defined strategies will remain committed to achieving the envisaged policy objectives, so investors can count on the existence and importance of these green technologies and on their follow-up development in 10 or 20 years.

In fact, much of the emissions reductions needed for achieving the environmental ambitions hinge on technologies that are still not commercially deployed or even at the large prototype or demonstration phase.[51] Infrastructure enabling the gradual deployment of these technologies in reduced scale to prove their technical feasibility and commercial viability is thus needed. Such pilot and demonstration facilities could mitigate much of the risk associated with full-scale implementation of a commercial facility. However, since running such facilities also usually requires considerable investment,[52] public support should be available to offset the first-mover disadvantage. While pilot facilities can often be financed from research funding, larger demonstration plants might have to operate at close to commercial conditions where financing opportunities are more scattered. Therefore, it should be considered to develop facilities that can operate on a longer term, beyond the purpose of merely demonstrating the operation of a certain technology.[53] Indeed, demonstration facilities not only help with developing the technology towards market readiness but can also be used to optimize already commercialised technologies or to help to further integrate it within existing systems.

Since green technologies typically function in ecosystems, they often need available sustainable resources, transportation methods, enabling infrastructure and operating third parties across the value chain to viably function. These new technologies can often viably function in an ecosystem, which requires that available sustainable resources, transportation methods and service infrastructure and third parties consuming or processing the final products be in place. Many of these demonstration projects are standalone projects issuing financial instrument such as bonds to finance the project or receiver of project funding from public funds. This makes the investors concerned as it is unclear how these projects will remain viable after the public resources dry up. Therefore, there is a need not just for a better communication but better coordination of these projects and their public financing in order ensure that complementary projects emerge across the value chain, which would create a viable ecosystem both from supply and demand perspective, leading to a much clearer business case for investors. This would require smart blended financing, which strengthens the synergies between the different regional, national and EU funds and uses them in a complementary way. Thus, financing would be ensured from early stage until first years' operating costs with public funding are being phased out and replaced by private capital and public loan guarantees gradually.

If these risks are being addressed, it is important to gather the most available data on the operating projects as often there is no historical data available for green technologies. EU reporting guidelines could facilitate the creation of data rich business cases. This data would need to be built in in new risk assessment models particularly for the use of institutional investors. Policy makers and industry should work together to make the most data available for risk assessment purposes and could develop new risk models to factor in non-financial risks of green technologies. Better understanding of the business case, attached risks and underlying data could facilitate crowding in private investment.

Lastly, despite the fragmented market of IP assets and green innovation, matching potential investors with investees has not been confirmed as a major barrier. However, matching investors with mature start-ups and SMEs developing mature products has been identified as a challenge. Start-ups are often considered as immature with unclear financial structure even if their development could have a promising market potential. Even if the start-up is mature and owns a green technology with good growth potential, it is often the case that they are unaware or reluctant to discover private financing such as commercial loans. Therefore, it is essential to develop the business management –including IP asset management- and financial planning literacy of start-ups and SMEs and raise their awareness of private and public-private financing opportunities.

4.4 Other political and regulatory aspects

Potential barriers to technology transfer: Political and regulatory aspects

- Over-complexity or fragmentation of the regulatory framework at EU-level / within the Member States;
- Lack of regulatory certainty for new, emerging technologies;
- Limitations of the European Research Area (ERA);
- Unused potential for synergies: Lack of a coordinated approach on technology transfer at EU-level (Member States implement similar, yet diverging national technology transfer policies);
- Horizontal approach to technology transfer is unable to take potential sectoral needs (*e.g.* as regards green technologies) into account;
- Heterogeneous diffusion and effectiveness of technology transfer concepts and structures across the EU and its RTOs;
- Contradictory policy interventions (*e.g.* fostering of technology transfer vs. efforts to keep talented researchers in academia) or shifting policy priorities endanger long-term planning / investments;
- Lack of effective incentives in the research environment to focus on technology transfer instead of a more traditional academic career path;
- Political and historical evolution of research institutes in EU Member States may not favour entrepreneurship.

Both financial and legal issues related to technology transfer result from the regulatory framework in place. However, there are also regulatory aspects related to hard and especially soft policy instruments that are not covered in the previous sections on finance and intellectual property. In addition, the cohesion of the overall framework – covering all policy instruments relevant to technology transfer – is a principal factor. In this regard, soft policy instruments are key facilitators for technology transfer and play a vital role in improving performance and binding the various aspects of technology transfer together. A financial and regulatory framework is unlikely to succeed without a range of measures connecting the worlds of industry, finance and research. In this regard, different elements and details of the policy framework can be potential barriers to technology transfer and commercialisation. The box at the beginning of this section sets out several potential barriers to technology transfer linked to such political and regulatory aspects.

While Technology Transfer policy has been gradually developed at EU-level with the development of the European Research Area (ERA), performance issues in comparison with other international competitors persist (in particular the US and China). Most notably, the European paradox of a high level of scientific output, being converted into a comparatively lower level of innovative products and services still holds true.

Besides providing funding for research and innovation, the EU approach to technology transfer has mostly focussed on soft policy instruments. Such "soft" instruments range from outreach activities, network building, stakeholder roadmaps, training programmes, guidelines, advisory services and the exchange of best practices. Given the status of research and technological development as a shared competence between the EU and its Member States, the EU has been particularly active in many of the aforementioned areas (see section 3.2). Bringing all Member States and all relevant stakeholders to the table as well as sharing best practices is a clear EU added value in many areas, certainly also in technology transfer. Notwithstanding the value of EU initiatives, EU actions are only complementary to national policies and therefore limited to a certain degree.

The specificities of the multilevel governance system of the EU, as well as the status of research and technological development as a shared competence, results in a wide range of potential barriers to technology transfer. These are linked to the inherent complexity of such a system. We also have to keep in mind that knowledge/technology transfer initiatives and instruments have only gradually been developed over the last two decades at EU-level.

Besides the hard and soft policy-instruments in place, the general political context and agenda can have an impact on technology transfer. The political context can determine the level of financial support, prioritise certain sectors and set objectives (*cf.* European Green Deal). For example, political priorities can have a strong impact on certain technologies, such as for example the consequences of the German energy shift towards renewable energy. In this regard, it can be expected that the European Green Deal and its follow-up might have a positive impact on green technology transfer.

Following this logic, the political response to the COVID-19 crisis will have direct and indirect repercussions on technology transfer. For example, support for R&D&I, including technology transfer, could suffer, as political priorities shift to direct crisis relief. Moreover, many private companies may invest less in innovation. Investors could tend to reduce risks and invest less in emerging technologies. This may also have to be factored into the EU policy response.

5. CASE STUDIES

The following section presents four selected case study areas for this report: Hydrogen technology, batteries, artificial intelligence and carbon capture, utilisation and storage technologies.

All four areas represent green technologies that are expected to play a key role in realising the ambition of the European Green Deal to achieve a climate neutral EU economy by 2050. The case studies are based on expert and stakeholder interviews, conferences and seminars, as well as complementary desk research.

5.1 Case study: Hydrogen technology

In short

- Hydrogen technologies, and in particular green hydrogen generated from renewable energy sources, have emerged as one of the key enabling technologies for the transition to a zero-emission society. As recent European and national hydrogen strategies highlight, there is a political impetus for further accelerating the deployment of hydrogen. EU-level initiatives, such as the Fuel Cells and Hydrogen Joint Undertaking or the newly launched European Clean Hydrogen Alliance, have been and will be crucial to boost the technology transfer and commercialisation of hydrogen technology. In this regard, European policy makers need to support the rapid scale-up of hydrogen production with a view to lowering the costs of green hydrogen and electrolysers to competitive levels. Further public action will be required to establish adequate infrastructures and to provide a fit-for-purpose regulatory environment. In addition, the public sector may need to help build up the end-user demand for green hydrogen and sustain the next generation of hydrogen innovation in its early stages. ▪▪

Many experts consider hydrogen to be one of the most promising energy carriers as regards the transition to a climate neutral economy. Generated from water by electrolysis, hydrogen can be used as a zero-emission power source and industrial feedstock, emitting only water and heat. While it is currently still produced mostly from natural gas ("grey hydrogen"), the potential lies in zero-emission "green hydrogen" made from renewable energy, such as solar or wind energy. "Blue hydrogen", where emissions resulting from the production are largely captured, stored or reused, is often put forward as a potential intermediate solution. In the terminology of the EU Hydrogen Strategy, the latter is called low-carbon hydrogen, while green hydrogen is referred to as renewable or clean hydrogen.

According to the ambitious long-term visions of hydrogen stakeholder associations, hydrogen could cover 18% of the total global energy demand in 2050.[54] In Europe, 2,250TWh of hydrogen could be generated by 2050, accounting for about a quarter of the EU's total energy demand. Already by 2030, this optimistic scenario forecasts a European market of about EUR 85 billion, of which a majority could be served by European companies, adding hundreds of thousands of jobs.[55] All these scenarios currently remain little more than informed speculation but is undoubted that the successful build-up of a clean hydrogen industry in Europe would bring about substantial environmental and economic gains.

Interest in hydrogen technology and its various technological applications has varied over the last decades (*e.g.* there have been discussion on the use of hydrogen in transport already in the 1980s). Having had its ups and downs, the increased pressure to drastically reduce our societies carbon emissions has recently led to a new peak in the attention on hydrogen as an energy carrier and industrial feedstock for a wide range of potential uses. Mature hydrogen technology is readily available for deployment, for example in industrial production (*e.g.* in oil refining, ammonia, methanol and steel production) and in transport (for passenger vehicles and in particular for heavy-duty vehicles).

Many experts see a key role for hydrogen as a feedstock to decarbonise industry, in particular the so-called hard-to-abate sectors (*e.g.* chemicals, metal industry). In transport, the role of hydrogen vis-à-vis battery-based solutions remains debated for passenger vehicles (where batteries seem to lead the race) with stronger use cases in heavy-duty vehicles and the transport of goods (with potential use in rail and maritime transport). In addition, hydrogen is considered a promising energy storage solution to balance out peaks in the generation of renewable energy. Many other use cases are imaginable, such as heating for buildings. This diversity in use cases highlights that speaking of "hydrogen technology" is a generalising term. However, all use cases require substantial amounts of hydrogen as energy source and are thereby based on hydrogen production technology. While the potential is clear, for its break-through hydrogen will need a massive scale-up in production, infrastructure and use. The next 10 to 25 years are currently seen as the decisive moment for the technology "to make it or break it". With a new array of hydrogen projects recently launched or announced to be launched soon (*e.g.* investment pipeline of the European Clean Hydrogen Alliance), the coming years will be decisive for the technology to deliver on the expectations of large-scale deployment and a shift towards green hydrogen.

At the same time, technological challenges remain and solutions based on green hydrogen are not yet widely deployed. Further technological progress should enable a more efficient electrolysis, new types of electrolysers (such as Solid Oxide Electrolysis Cells), cost reductions, a higher durability of components and in particular better transport and storage solutions. In this regard, innovation and technology transfer will be a key factor to enable the wider uptake of clean hydrogen across different sectors in Europe.

According to Hydrogen Europe's Clean Hydrogen Monitor report, about 450 hydrogen production sites were in operation in Europe at the end of 2018. The consumption of hydrogen currently remains concentrated in just four countries, with a joint consumption of more than 50% of the total: Germany (22%), the Netherlands (14%), Poland (9%) and Belgium (7%). Without factoring in by-product hydrogen (generated by certain industrial processes), the total European hydrogen production capacity in 2018 added up to 9.9 Mt of hydrogen per year (27 thousand MTD). As a relevant indicator, one should note that over 92% of all hydrogen production was based on fossil fuels. The share of low-carbon, blue hydrogen production from fossil fuels with the use of CCS/CCU technology played an insignificant role at around 0.7% of the total production.[56] Electrolysis, the electricity-based generation of hydrogen remains for now equally negligible and is estimated to be around 1% of the total production.[57] As a result, the current availability of low-carbon hydrogen is very limited. The current demand for hydrogen is driven largely by industrial applications, in particular refineries (45% of total hydrogen use), followed by the ammonia industry (34% of total hydrogen use) and chemical industry (12% of total hydrogen use).[58]

EU policy approach

Given its potential to reduce carbon emissions, hydrogen technology has been attracting considerable attention at EU-level over the last years. The European Green Deal mentioned clean hydrogen and fuel cells as a priority area to achieve a zero-carbon economy. Moreover, the new Industrial Strategy for Europe announced a new Industrial Alliance on Clean Hydrogen.

However, already prior to the European Green Deal, hydrogen was a focus area at EU-level. Originating from the High Level Group on Hydrogen launched in 2002, the European Hydrogen and Fuel Cell Technology Platform (2004-2007) under the Framework Programme for Research led finally to the creation of the Fuel Cells and Hydrogen Joint Undertaking as a as a public-private partnership in 2008. Under the Horizon 2020 Framework Programme, the Fuel Cells and Hydrogen Joint Undertaking received a total funding of about EUR 1,33 billion for the 7-year funding period from 2014 to 2020.[59]

The Fuel Cells and Hydrogen Joint Undertaking functions as a public private partnership supporting RTD activities in fuel cell and hydrogen technologies in Europe. Within the Fuel Cells and Hydrogen Joint Undertaking, the main partners are the European Commission, Hydrogen Europe (representing the fuel cell and hydrogen industries) and Hydrogen Europe Research (representing the research institutions). Meant to speed up the market introduction of hydrogen technologies, the Fuel Cells and Hydrogen Joint Undertaking's main objective is to facilitate technology transfer and commercialisation for the technology by pooling private and public resources.

The Strategic Forum for Important Projects of Common European Interest, set up by the Commission in 2018, analysed the needs to further develop a range of strategic value chains, including hydrogen technologies and systems. In its final report, the Strategic Forum outlined its vision for the future development of hydrogen in Europe.[60]

In July 2020, the Commission's Hydrogen Strategy for a Climate-neutral Europe put forward a roadmap to boost the widespread deployment of hydrogen in the EU, setting out an agenda of financial and regulatory actions to support the roll-out of hydrogen technologies.[61] The Strategy sets out deployment targets in three phases:

1. 2020 – 2024: installation of at least 6GW of renewable hydrogen electrolysers in the EU and the production of up to 1 million tonnes of renewable hydrogen with a focus on decarbonising existing hydrogen production and facilitating the take up of hydrogen in new end-use applications (in particular industrial and processes and heavy-duty transport).
2. 2025 – 2030: at least 40GW of renewable hydrogen electrolysers and EU production of up to 10 million tonnes of renewable hydrogen with a view on making intrinsic part of the integrated EU energy system.
3. 2030 – 2050: large-scale deployment of renewable hydrogen technologies across all hard-to-decarbonise sectors.

Jointly with the Hydrogen Strategy, the European Clean Hydrogen Alliance was launched. The aim of the Alliance is to bring together all main stakeholders to build up a strong pipeline of investments in hydrogen projects, following up on the deployment objectives set by the Hydrogen Strategy. In respect to the case study on battery technology, it is relevant to remark that the Alliance explicitly aims to follow the model of the European Battery Alliance, which is seen as a success. In addition, the Commission's recovery plan, the funding instruments of Next Generation EU, including the Strategic European Investment Window of the InvestEU programme and the ETS Innovation Fund, are put forward in the Hydrogen Strategy to ensure financial support in the short- and mid-term and to bridge the investment gap for renewables related to the COVID-19 crisis.

Reflecting the need for a holistic view on the broad integration of hydrogen in the European energy system, the Hydrogen Strategy was released jointly with the Commission's new EU Strategy for Energy System Integration. Furthermore, any future strategy on clean steel is very likely to feature hydrogen at its core.

Overall, 2020 marked a year of strategic orientation for hydrogen, emphasising the high hope of public and private stakeholders on the technology.[62] The coming years will need to live up to these objectives and pave the way for the deployment of hydrogen technology. This deployment is likely to be facilitated by a so-called Important Project of Common European Interest (IPCEI) on clean hydrogen. Furthermore, the Fuel Cells and Hydrogen Joint Undertaking is set to continue as the European Partnership for Clean Hydrogen under the future Horizon Europe Research and Innovation Programme (2021-2027) with a revised scope and a wider range of partners.[63]

Technology transfer and commercialisation of hydrogen technology

Hydrogen seems to be on the verge of becoming a central part of tomorrow's energy system reflecting the potential contribution of renewable hydrogen to curb emissions. It is Europe's ambition to be the global leader on hydrogen technologies, in particular as regards green hydrogen solutions.

Currently, most European experts seem to agree that the EU for now is very well positioned as regards hydrogen technology vis-à-vis its global competitors. Nonetheless, hydrogen technology is promoted and researched globally and other markets are competing with European technology and companies for the lead. While a drastic loss in EU capacities vis-à-vis cheaper Asian competitors (as occurred in photovoltaic power and in batteries) is currently considered not an imminent threat, in particular China is likely to become a major global competitor, not the least on the price based on its large-scale production capacities. The United States, Canada, South Korea, Japan and Australia are also well positioned as regards hydrogen technology. This can be seen as a further call for Europe to increase its production fast to build up EU strong value chains.[64]

As the previous section has shown, there has been a lot of action at EU-level on the technology with a view to supporting its uptake. This attention seems to have paid off already, even if recent mid- and long-term strategies still need to show their ability to realise their ambitious objectives. A systematic approach seems to be certainly needed to start the gradually build-up of hydrogen in the EU's energy system. In this regard, risks remain that national hydrogen strategies could compete with each other instead of ensuring complementarity among Member States. In addition, the interest in and the research on hydrogen remain heterogeneous across the EU. While many Member States are supporting hydrogen research and have their own national deployment strategies, there is further room for mainstreaming hydrogen solutions in the Union and ensuring all Member States are in step on its deployment.

However, the European Green Deal, accompanying strategies and regulatory initiatives such as the European Climate Law help drive investments to technologies such as hydrogen as increasingly large-scale investors shift investments to green technologies.

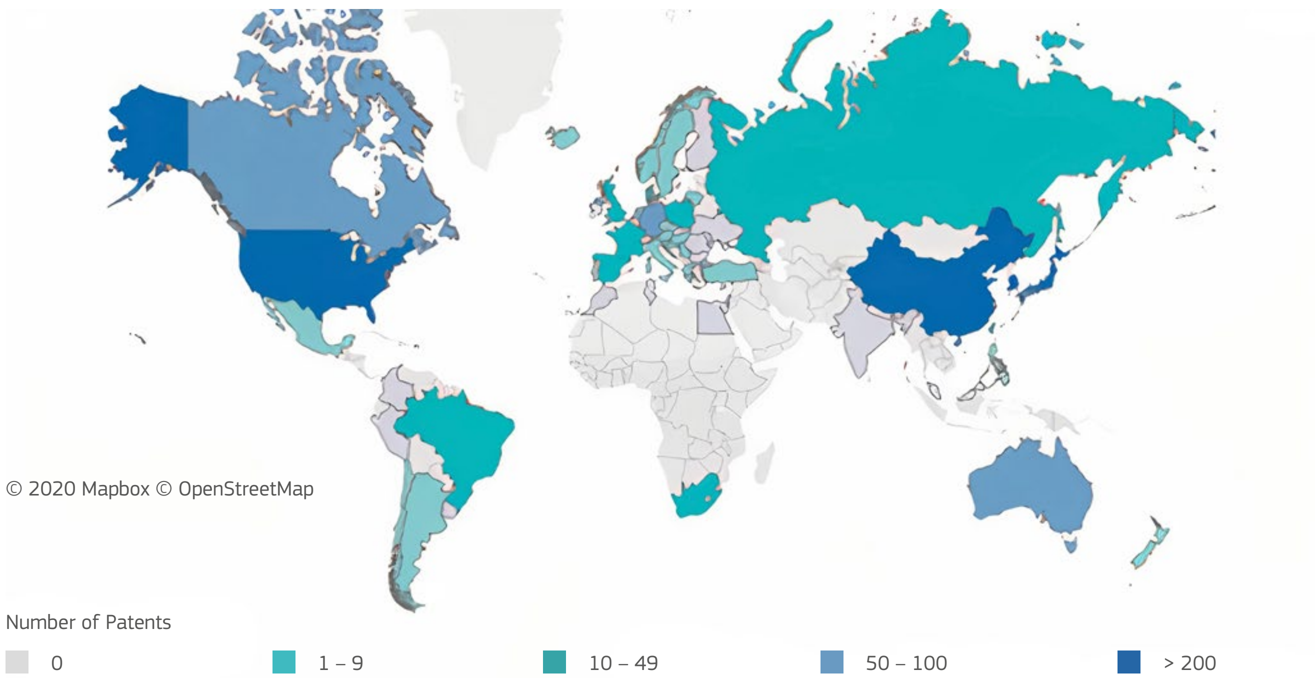


Figure 5: Number of hydrogen technology patents per country in 2019. Source: Fuel Cells and Hydrogen Observatory 2020.[65]

A strong political narrative helps to build trust in new technologies and incentivises investments in new zero-emission technologies, even if costs may be initially higher compared to established technologies.

Investment environment

A strong shift towards green investment helps in particular projects based on mature technology to secure funding. With the European Investment Bank (EIB) prioritising sustainable investments in its role as the "EU climate bank", new financing opportunities arise also for hydrogen projects.[66] The EIB's Climate Bank Roadmap 2021-2025 recognises "the need to increase investment in innovative green technologies – from early-stage research through to pilot demonstration of technologies, complemented with support for new business models" and "the importance of driving down the long-term cost of capital in capital-intensive green infrastructure"[67]. In addition, the EIB Roadmap links itself to the EU Hydrogen Strategy and stresses the need to deploy low-carbon hydrogen. For this purpose, the EIB intends to develop specific instruments to support the roll out of low-carbon hydrogen (*e.g.* carbon contracts for difference incentivising industries to switch to low-carbon hydrogen).

The strong involvement and commitment of resources from the public sector also helps private actors and financial intermediaries to limit the risk they take by investing or transitioning in clean hydrogen technology. Overall, the ever-rising need and interest in zero-emission technologies, in combination with clear policy signals, helps to create a market-pull effect for hydrogen technologies. As a prioritised research area, hydrogen projects have an advantage in finding interested investors or public funding. Nevertheless, especially early-stage technologies are likely to encounter difficulties when seeking access to risk capital.

While there is a strong and growing interest to finance hydrogen technologies, the access to venture capital continues to be a notable barrier for the technology transfer and commercialisation process. However, this barrier is perceived more strongly on the side of RTOs, while the industry perceives this as less of an issue in the current environment. In this context, the role of research institutes, universities and their technology transfer offices is crucial to pave the way for technologies out of the lab.

Most market actors, from banks to private companies, have a strong tendency to limit risk exposure when making investment choices, such as they are inherent to investments in new technologies. As with other potentially ground-breaking technologies, the vision of green hydrogen today exists mostly on paper. Many investors still need to know-how trust and acceptance in hydrogen-based solutions. While investments in hydro-gen have reached new highs, access to finance contin-ues to be a barrier in the technology transfer process.

With the market looking to invest in fully developed, mature technology, it continues to be difficult for projects at lower TRL[68] levels to source the desired funding. Where private actors are not willing or able to sustain technological innovation in hydrogen, there continues to be a gap which public funding may need to fill. In this regard, stakeholders, who see this as a clear-cut case for public sector support, consider the fostering of high TRL levels based on public invest-ments necessary.

Role of EU level cooperation

Over the course of its activity, the Fuel Cells and Hydrogen Joint Undertaking served as a major source of funding for hydrogen technology projects at various TRL levels, matching public and private contributions. Since 2014, the Joint Undertaking has funded 91 projects with a total financial contribution of EUR 460.9 million.[69] In 2019, for every euro contribution from the EU budget to the signed grant agreements of the Joint Undertaking, the stakeholder organised within Hydrogen Europe Industry and Hydrogen Europe Research committed to spend about one and a half euro either on the projects of the Joint Undertaking or on additional activities.[70]

Just as importantly, since its establishment the Fuel Cells and Hydrogen Joint Undertaking has served as a forum to connect industry, researchers and policy makers at EU-level, resulting in an immeasurable networking effect. Amongst others, such networking may help launch new cooperation and to better align research priorities with the needs of industry. In this regard, the Fuel Cells and Hydrogen Joint Undertaking serves as a clear example for the added value of streamlined EU-level cooperation and pooling of resources.

The newly launched European Clean Hydrogen Alliance should act as a complementing factor to ensure further networking gains. In addition, it should help realise new investments in hydrogen. In particular, the Alliance aims to realise the installation of at least 6 GWh of renewable hydrogen electrolysers in the EU by 2024 and the installation of 40 GWh of renewable hydrogen electrolysers by 2030 (following the objectives set in the EU Hydrogen Strategy).[71]

Renewable energies and infrastructure as enabling factors

As recognised in the EU Hydrogen Strategy, the access to renewable energy is a major facilitator for the generation of green hydrogen. In this regard, hydrogen advocates argue strongly against any regulation that might disincentivise the use of renewable energy capacities for this purpose. As abundant and cheap renewable energy is a pre-condition for the generation of green hydrogen at a low cost, an accelerated transition to renewable energy is a strong facilitator for hydrogen. Regulations should not disincentivise the utilisation of renewable energy to generate clean hydrogen.

Deploying hydrogen at a large-scale across sectors has a strong infrastructure component, given hydrogen needs a fit-for-purpose infrastructure to allow for its delivery to the end-user. Since hydrogen is similar to natural gas, repurposing the existing gas grid infrastructure should be able to fill some of the infrastructure needs.[72] However, there will also be a need for investment in new hydrogen infrastructure. Hydrogen technologies will require, amongst others, changes to existing gas pipelines and potentially also new pipelines, as well as usage-specific infrastructure such as a network of hydrogen fuelling stations for heavy-duty transport. Building up such infrastructure from the ground will most likely require public intervention and investment, as it is unlikely that all necessary infrastructures would be build-up by the industry itself.

Regulatory environment

European policy makers will need to ensure that the regulation on common technical and safety standards concerning hydrogen-related infrastructures is adequate to support the rollout of hydrogen. An enabling, stable regulatory environment is particularly important to enable large-scale projects and investments. In this regard, the Strategic Forum for Important Projects of Common European Interest called for a level playing field between fossil-based solutions and hydrogen-powered alternatives as well as a reform of relevant regulations and standards that will support the objective of a swift technological transition to hydrogen.[73]

The EU Hydrogen Strategy reflects this understanding. Among the EU regulations considered relevant for the recent EU Hydrogen Strategy are notably the Renewable Energy Directive, the Emission Trading System, the Alternative Fuels Infrastructure Directive as well as the internal energy market and gas market regulation.

Nevertheless, stakeholders remain concerned regarding the speed of regulatory adjustments or their adequateness. To enable the deployment of green hydrogen, the regulatory framework needs to lay out an enabling environment providing the regulatory certainty required for large-scale investment in long-term projects.

In detail: The HyLAW project

The HyLAW project is an interesting case for the cooperation of private stakeholders to achieve common goals in the deployment of a novel technology:

"HyLaw stands for Hydrogen Law and removal of legal barriers to the deployment of fuel cells and hydrogen applications. It is a flagship project aimed at boosting the market uptake of hydrogen and fuel cell technologies providing market developers with a clear view of the applicable regulations whilst calling the attention of policy makers on legal barriers to be removed.

The project brings together 23 partners from Austria, Belgium, Bulgaria, Denmark, Finland, France, Germany, Hungary, Italy, Latvia, Norway, Poland, Romania, Spain, Sweden, Portugal, the Netherlands and United Kingdom and is coordinated by Hydrogen Europe.

The HyLaw partners will first identify the legislation and regulations relevant to fuel cell and hydrogen applications and legal barriers to their commercialisation. They will then provide public authorities with country specific benchmarks and recommendations on how to remove these barriers." [74]

Scaling up the European hydrogen industry

While there are many reasons to be optimistic about the prospects of hydrogen technology, important barriers persist, in particular for the most ambitious scenario of hydrogen utilisation. While hydrogen has become a buzzword and there are many reports highlighting its potential, current numbers highlight that the way to a systemic change remains long. The current production capacities for clean hydrogen remain limited and are mostly small-scale projects.[75]

In its 2019 report, the Strategic Forum for Important Projects of Common European Interest diagnosed a supply/demand deadlock as the major barrier for hydrogen to overcome.[76] Economies of scale would allow achieving the required cost reductions to make hydrogen solutions competitive in the market.

The competitive cost of green hydrogen compared to fossil alternatives and in particular, grey hydrogen is assumed to be the key factor for its success. Accordingly, the current cost gap is considered an important barrier to overcome. Overcoming this gap, would require on the one hand cheap, mass produced electrolysers to drive down CAPEX in green hydrogen projects. On the other hand, a stable demand for green hydrogen is needed.

Estimates by Bloomberg New Energy Finance expect that a strong scale up of clean hydrogen could result in a benchmark cost for hydrogen of USD 2/kg in 2030 and USD 1/kg in 2050, allowing green hydrogen to be competitive in most locations.[77] The required scale-up is likely to require public support and policies aiming at the promotion of green hydrogen technologies made in Europe. Green hydrogen will need to become cheap to become an automatic choice. In this regard, regulatory incentivise (including the price for carbon emissions) play an important role in setting determining the viability of industrial investments.

At political level, a stable narrative on policy priorities will help to convince investors and the wider public of the long-term prospects of clean hydrogen. Moreover, policy makers need to reduce uncertainties related to the future regulatory environment.

As a straightforward solution to increase the market for hydrogen while reducing carbon emissions, hydrogen stakeholders call for allowing the feed in of hydrogen into European natural gas grids. Such a blending in of hydrogen into natural gas grids seems to be technically feasible up to 10% (with varying estimates).[78]

A scale-up would be based on mature hydrogen technologies readily available today. However, this commercialisation of mature technology would lay the ground for future of generations of hydrogen-based technologies. In this regard, it would likely have a strong indirect impact on the long-term perspective of the technology in the field

If hydrogen were indeed to become a major pillar of the European energy system, the technology transfer and commercialisation of green hydrogen innovation would be facilitated by a strong market-pull and competition for better hydrogen technology.

Research priorities

While a substantial scale-up needs to be pursued to create economies of scale to drive down costs, R&D efforts will be required to improve hydrogen further and enable its long-term success. The public sector needs to ensure that RTOs have sufficient resources and funding streams to remain global leaders in hydrogen R&D. In addition, the policy framework should set priorities and provide a stable regulatory environment. Public funding and regulation should aim at de-risking private investments and attracting private investments. The International Energy Agency (IEA) highlights in particular the following priorities:[79]

- Well-developed technologies at high TRL levels (*e.g.* fuel cells and electrolysers) could benefit from R&D efforts oriented at the improvement of their performance and manufacturing processes, thereby lowing costs and increasing their operational efficiency and lifetime.
- Less-developed technologies at lower TRL levels (*e.g.* hydrogen-derived fuels, further end-use cases) should be supported financially, in particular to demonstrate their viability to attract investors by building trust, lowering risks and driving down financing costs.
- Least-developed technologies, still relatively far from commercial viability (*e.g.* biomass gasification with CCUS and seawater electrolysis), should receive R&D support to ensure progress as regards the validation process with a view on future commercialisation. Many of these technologies will be key to materialise the high potential of hydrogen technology.

Conclusions: Case study on hydrogen technology

Hydrogen has emerged as one of the key enabling technologies for the transition to a zero-emission society. Accordingly, the technology transfer and commercialisation of clean hydrogen technology have the potential to contribute substantially to the objectives of the European Green Deal.

In 2019, the IEA concluded, "clean hydrogen is currently enjoying unprecedented political and business momentum, with the number of policies and projects around the world expanding rapidly."[80] At the same time, the agency called for a push to scale-up the hydrogen industry technologies in view of lowering costs to ensure widespread use. In the meantime, important political progress has been achieved at EU-level. With the EU Hydrogen Strategy, national hydrogen strategies and a strong bundling of industry and research stakeholders at EU-level, the prerequisites for the success of hydrogen technology have been established.

At present, stakeholders seem to have an optimistic outlook on the future of hydrogen technology. However, the European Commission and national governments now need to follow-up on these plans to ensure rapid progress on these ambitious. The main challenge remains the exponential scale up of green hydrogen and the systemic change of the European energy system. While the primary objective is to curb carbon emissions, the deployment of hydrogen technology is also a global industrial competition for a technology that is likely to be an important part of future energy systems.

In the immediate future, the scale-up of clean hydrogen production capacities and infrastructure will need to be prioritised. This means rapidly expanding the production of electrolysers in the EU, to enable the generation of green hydrogen. Furthermore, it requires the development of a corresponding market for green hydrogen in end-uses. In this respect, policy makers may need to prioritise certain end uses.

At EU- and national level, the regulatory framework needs to enable the deployment of clean hydrogen, ensuring regulatory certainty and stability. Public actors will also need to address market failures, for example by guaranteeing a demand for green hydrogen. Across the EU, common norms and safety standards should be provided to avoid any regulatory fragmentation. The public funding and policy support for R&D, in particular at lower TRL levels, will need to sustain the next generation of hydrogen innovation.

5.2 Case study: Batteries

In short

- As in the case of hydrogen, stakeholders in the battery industry are cautiously optimistic about the future, as the EU has made a clear political commitment to bring the battery cell production to Europe. Significant results have already been achieved through intensive coordination between R&I and the industry (*e.g.* European Battery Alliance). This collaborative ecosystem enables investment and supports knowledge transfer. Policy makers should be careful, however, to keep the opportunities open to novel battery technologies as there is a clear risk of directing all attention to the current winning technologies (Li-ion batteries). Risks remain high for innovative start-ups in the field and public support is still needed to reduce the financial risk to investors and to foster integration along the whole value chain. Overall, the battery industry may, however, serve as a good example for the deployment of other green technologies.

Storing energy in batteries is far from a new concept, yet batteries belong to one of the most researched technologies nowadays. In principle, a battery consists of two electrodes isolated by a separator and placed in electrolyte to promote the movement of ions. There are, however, numerous variations to this basic concept catered to applications. While first lead-acid batteries were developed more than 100 years ago, they are still used for providing motive power.[81]

Lithium-ion (*"Li-ion"*) batteries, first developed in the 1980s and commercialised in 1991 for use in consumer electronics, on the other hand, are widely used for consumer electronics, electric mobility and have large potential for stationary power storage. The demand for Li-ion batteries as the energy source for electric vehicles (EVs) is expected to continue growing exponentially in the coming years as the automotive sector will shift towards more carbon-neutral solutions.

Batteries will also be crucial to smoothing out the inherent intermittency of renewable energy in the energy grid, ensuring uninterrupted power supply and enabling efficient managing of energy system.

This makes batteries, and specifically Li-ion batteries, a strategic value chain and one of the key green technologies to achieve European Green Deal ambitions, in particular as an enabler for the electrification of our societies. Yet not long ago, Li-ion battery manufacturing was practically non-existent in the EU. Around 80% of global battery cell production capacity is currently in Asia. Importing battery cells from Asia increases the carbon footprint of electric vehicles and shifts the value creation and jobs away from Europe. As batteries represent about 40% of the value of electric vehicles, a reliable supply of EV batteries at a reasonable price will become a pre-condition for automotive sector to remain competitive. As COVID-19 crisis has shown, outsourcing key technological products through long supply chains may imply vulnerabilities for the industry.

To prevent the EU's technological dependence on its competitors and to capitalise on rapidly increasing demand for batteries[82], it became an EU political priority to supercharge the European battery industry. Since 2017, public investment into developing the European battery value chain and building production facilities in Europe was ramped up to an unprecedented level.[83] Since then, Europe continues to move at pace to create a battery industry that will supply the continent with batteries that are both sustainable and competitive.[84] The European commitment to decarbonisation ensures an ongoing market demand for batteries in Europe. The challenge therefore is to safeguard supply and to keep up with the fast pace of developments in the field by supporting both innovation itself and the deployment of new technology.[85]

While most of the attention is on creating production capacity of Li-ion battery cells in Europe, other aspects of the battery value chain, such as the supply of raw materials and end-of-life disposal, recycling or repurposing of batteries are also in the spotlight.

EU policy approach

The first step towards establishing the European Union's policy concerning batteries was made in 2008 in European Strategic Energy and Technology Plan[86]. A decade later, battery technology features in all major strategies for achieving the European Green Deal ambitions.[87] The European Green Deal emphasises the essential role of collaboration among stakeholders, as well as other initiatives leading to alliances and to a large-scale pooling of resources.

Indeed, cooperative platforms and alliances significantly facilitate the coordination among all stakeholders in the European battery ecosystem. In 2017, the **European Battery Alliance**[88] was launched, marking the beginning of the battery's place at the forefront of the EU's policy interest. The Alliance aims at building an independent, sustainable and resilient battery industry in Europe and supporting the scaling up of innovative solutions and manufacturing capacity. The Alliance is helping to foster cooperation between stakeholders across the value chain and design measures to kick start battery production in Europe.

The European Battery Alliance was shortly followed by other EU-level initiatives to support research, innovation and deployment of battery innovation. The European Technology and Innovation Platform (ETIP) on batteries, **BatteRIes Europe**[89], is a platform that brings together industrial stakeholders, the research community and EU Member States to advance battery research priorities and foster cooperation between relevant battery research programmes at EU and national levels, as well as private research initiatives. It monitors and coordinates the research on the next-generation battery technologies, across all TRL levels, stages of the value chain and battery applications.

To prepare the commercialisation of the next-generation battery technologies, a new Battery Partnership[90] is in preparation. The Partnership aims to accelerate the development, industrialisation and deployment of various battery technologies across the whole value chain (including processing raw materials, innovative battery cells, packs and systems, and end-of-life management) in order to bring Europe ahead in the race to manufacture the future generation of sustainable batteries. Stakeholders will coordinate development along the technology readiness level scale (TRL). The Partnership's focus will be on close-to-market Li-ion technologies (TRL 5-8), as well as on technological solutions that are less mature (TRL 2-4) but hold longer term promise, such as solid-state batteries.

This cooperative European battery ecosystem illustrates well how policy efforts at bringing stakeholders together may close the gaps in technology transfer. In Europe over the last 4 years many informal contacts have been built and R&I projects between the research and industry have started. Battery research and battery industry are no longer distinct, scarcely interacting worlds where new promising technologies remain only on the pages of scientific journals. The current and future market needs are closely reflected in university research.

With regard to the financing of battery research and deployment, the EU budget is already providing important funding opportunities to support battery research and innovation.[91]

In the new Multiannual Financial Framework, a new InvestEU will mobilise funding and investment, integrating several existing financial instruments. InvestEU's Innovation Fund should provide financing to support pre-commercial demonstration projects in low carbon technologies, including energy storage.[92]

Further public financial support for battery innovation is available through European Investment Bank and EIT Innoenergy innovation public-private partnership (a Knowledge and Innovation Community of the European Institute of Innovation and Technology).

The European Investment Bank (EIB) supports first-of-a-kind commercial-scale energy demonstration projects by providing them with loans, guarantees and equity-type funding. In particular, the EIB´s Innovfin Energy Demonstration Programme[93] helps to facilitate the demonstration phase of innovative energy projects, including battery pilot lines. The EIB expects to increase its support for battery-related projects along the entire value chain to more than EUR 1 billion of financing in 2020. This would match the level of support the EIB has offered over the last decade.[94]

In detail: Northvolt

In 2017, the European Investment Bank provided a loan of EUR 52.2 million through the Innovfin Energy Demo Projects facility to Northvolt gigafactory start-up in order to create a demonstration plant for the manufacturing of Li-ion battery cells.

To build the first gigafactory in Europe, Northvolt also received additional EUR 350 million of public funding through EIB's EFSI strategic investment scheme. This makes the Northvolt gigafactory the EIB's biggest investment in battery technology to date. In a textbook example of how public funding can help attract private investment, public financing has helped to raise a total of EUR 2.5 billion in equity and debt financing.

Northvolt based its business model on innovative manufacturing processes and vertical integration of the whole value chain from mining to recycling while utilising 100% of renewable energy in the process. On the recycling program, it cooperates with Chalmers University in Sweden. Northvolt has the ambition to build 150 GWh of advanced battery manufacturing facilities in Europe by 2030, covering 25% of market share in Europe.

EIT Innoenergy innovation public-private partnership drives innovation in battery technologies (among other) by fostering cooperation among companies, research labs and universities and by providing support, knowledge and financing opportunities for companies with innovative solutions. EIT invests in innovative start-ups and businesses and facilitates access to venture capital firms. Specifically in the field of battery innovation, EIT Innoenergy co-designed the **Business Platform for Investors95**, a platform to match investors in battery technology with those seeking financial backing. The Business Platform gathers frontloaded investment in battery innovation from public and private sources and helps to reduce investment risks and shorten the time to investment.[96]

Further financing for battery research and innovation projects is available at national and regional level or from private investors. Specifically in the field of batteries, the European Commission approved first wave of EUR 3.2 billion of national state aid funding for battery R&D&I and industrial deployment under the **'important project of common European interest'** (IPCEI) which allows EU Member States to fund the ground-breaking innovation. The project is expected to unlock a further EUR 5 billion in private investment. The second IPCEI, focused on funding battery pre-commercial activities, is currently being prepared.[97]

Ongoing battery research in Europe

The ever-increasing demand for batteries is mirrored in a boom in battery research.[98] The research of new battery technologies and chemistries is strongly supported by public research grants – whether the Horizon 2020 Programme or the future Horizon Europe, or national and regional programmes. This funding target projects across all technology readiness levels and chemistries with the objective to develop competitive batteries with a low carbon footprint and sustainable life cycle.

Given the general prominence of Li-ion batteries, it comes as no surprise that they are also the most researched battery technology[99] - whether in research of new cathode and anode materials, research to improve certain characteristics of the present Li-ion technology, or other developments at cell level. In general, many of the new developments for Li-ion batteries are in the optimisation of existing technology.

Numerous technology strands are under development to improve the Li-ion battery and counter the disadvantages of Li-ion batteries (including their lifespan, flammability and toxicity and resource scarcity). New generations of Li-ion battery technology are being developed. These new generations of Li-ion batteries might well be many years away from commercialisation but are already promising to improve the battery performance.

NEW LITHIUM–ION BATTERY CHEMISTRY	
GEN 3	Advanced litium–ion
GEN 4	All–solid–state lithium–ion, with conventional or lithium metal as an anode

Table 1: New Lithium-ion battery chemistries

While Li-ion batteries are currently the focus of attention, it is understood that Li-ion chemistry may once be replaced by other solutions using different electrochemical combinations. Energy storage field and various niche battery markets might be better served by batteries with different characteristics than those crucial for the automotive industry – *e.g.* by batteries with longer lifespans, longer storage duration, improved safety and shorter supply chains of components. However, the development and diffusion of these *"successor"* technologies will likely take decades.[100] It is nevertheless in Europe's interest not only to react to existing market needs but also to achieve competitive advantage by getting to the forefront of the development and deployment of the breakthrough batteries of the future. Public research plays a particularly strong role in developing long-term predictions for the future of battery technologies, as well as for early-stage innovation of such battery technologies.[101] Fundamental university research on batteries with self-healing properties to prolong battery life is also ongoing.

Europe's efforts in becoming self-sufficient across the battery value chain and making the whole Li-ion battery life cycle sustainable also encourage further research, for example in developing novel battery chemistries using raw materials present in Europe instead of rare materials that must be sourced from abroad.

Aside from new battery variations developed to avoid using certain minerals in battery production, the home supply of materials for battery production can also be achieved through the recycling. Due to the limited number of Li-ion batteries at their end of life now, innovation in recycling of Li-ion batteries has been limited.[102] However, battery waste is expected to grow because of the increased usage of battery-powered EVs and with it also the need to develop recycling and sorting technologies, as well as end-of-life and second-life applications.

While research on material sourcing and end-of-life processing of batteries is important, innovation largely focuses on the optimisation of the battery manufacturing process. This optimisation has led to savings in costs[103] and to energy efficiencies. As battery manufacturing process is heavy on energy consumption, optimisation of the manufacturing process is indeed essential.

Technology transfer and commercialisation of battery technology

It is clear that industry does much of the research in the battery sector itself. That is the case especially for research in battery application and optimisation of manufacturing and production.[104] However, for the purposes of the present study on the technology transfer of green technologies, it is the research done at universities and public research institutes that will be examined.

In fact, the Li-ion battery technology itself was developed as a result of a university research and is thus a case in point for successful technology transfer.[105]

While universities led the way in battery research a couple decades ago, currently universities and research organisations in Europe account only for 12.7% of international patent families in battery technologies. Most patents are attributed to the large companies[106] – a development that can be explained by the increasing maturity of the Li-ion battery technology.

While university research on Li-ion batteries is often overshadowed by industrial innovation, public research plays an undeniably prominent role in producing long-term projections in the battery field and advancing research along these long-term visions.

In detail: Battery 2030+

A strategically important university research project, **Battery 2030+**[107], looks at long-term research predictions in the field of batteries. Battery 2030+ will identify future research needs and fill the gaps for long-term research. Aside from predictions of the future developments at the cell level, the project also covers research needed for the development of smart functionalities of batteries.[108]

The research project will also aim to predict future revolutions in battery research itself that could be achieved using novel technologies. It is for example anticipated that the use of artificial intelligence in research can accelerate discovery of novel battery chemistries and material combinations more efficiently than the current trial-and-error approach.

Research of post-lithium batteries and other battery-related technologies at low TRLs is typically research conducted by public research institutes. According to the EPO, universities filed 21% of major patents in the field of research focused on alternative chemistries to Li-ion.

Universities and public research institutes also contribute to developing novel cathode chemistries, new generation of Li-ion batteries and other developments at the cell level. Major performance improvements may be unlocked by university research on the use of new disruptive technologies, such as nanotechnology or graphene in the battery cells technology.

Technology transfer

Given the demand for competitive battery technologies and the added value of university research results, it is essential to ensure a well-oiled transfer of new innovations from research institutes to the market.

By comparison with the other fields of green innovation, the typical lack of coordination among research and the market needs does not seem pertinent for batteries. Thanks to the platforms such as European Battery Alliance or Batteries Europe, various stakeholders of the battery landscape formally and informally collaborate, share knowledge and thereby bridge the gap between research and the market. A new initiative in making, Batteries Partnership, will also coordinate the development and commercialisation of the novel

technologies at lower TRL levels, further facilitating the technology transfer. It can thus be concluded that the *"silo"* mentality often seen in other green technology sectors is not typically a barrier to technology transfer in battery innovation in Europe. Even if the opportunities to network at fairs, conferences or meetings of stakeholder alliances may be currently limited due to the COVID-19, these limitations are perceived as temporary.

With regard to IP aspects of technology transfer in battery technology, it became clear from the interviews conducted that IP is not seen as a major bottleneck for battery technology transfer. First, the management of intellectual property in battery technology throughout the technology transfer process does not differ from practices in the transfer of other technologies. Moreover, the interviewed stakeholders shared the view that if there is a business case for a given battery technology, the industry usually manages to settle possible IP-related complications.

Commercialisation of battery technology

Similar to technology transfer, commercialisation of battery technology has also considerably improved in the recent years as batteries came into the public policy spotlight in Europe.

It comes as no surprise that in the recent wave of public support, much of the attention goes to the mature Li-ion technology that can be readily deployed and to the construction of battery giga-factories.[109] This focus on the manufacturing of mature Li-ion battery technologies, however, also helped to create a demand for innovations that have the prospect of improving the characteristics of Li-ion batteries, or bringing efficiencies to manufacturing, or ensuring ethical and self-sufficient provision of raw materials as part of the transition to a circular economy.

The strong policy push for competitive production of sustainable batteries in Europe provides a guarantee that there will be a business case for battery production in the long term. However, the European battery industry could have hardly been built without public financial support. Some of the public funding possibilities for battery commercialisation are described above in the section on EU policy framework. In particular, with regard to the commercialisation of battery technologies at lower TRL levels, EIB´s instrument Innovfin for financing demonstration projects is relevant and has already been successfully used in the field of batteries. Also EIT Innoenergy streamlines administrative help and funding opportunities for early-stage innovative projects in the field of batteries. Its Business Investment Platform helps matching investors interested in battery innovation with those seeking investment. Finally, informal cooperation of researchers with industry in various alliances helps to streamline the uptake and financing of early technologies.

Big industry players have not typically been willing to take up the business risk associated with the uptake of early-stage battery innovation. While there is no uncertainty about the future business case for batteries, the uncertainties relating to technology and its prospective success or failure seem to be a deterring factor there. The commercialisation of breakthrough innovation coming from research institutes is often a domain of start-ups which are willing to undergo greater business risks. These start-ups, however, face the usual problems with amassing sufficient finance to commercialise the technology. While public intervention – incentives, funding and investment – has helped considerably to de-risk private investment in the battery industry, private financing of battery innovation remains a challenge, especially for technologies of lower TRL levels.

In detail: HE3DA

The case of Czech company HE3DA[110] illustrates some of the difficulties faced by innovators when looking for private investment in order to scale up an innovative battery invention and set up industrial manufacturing.

The HE3DA (High Energy 3D Accumulator) is a sustainable lithium battery technology that uses nanoparticles to achieve improved battery characteristics. The invention, developed as a result of university research, is patented worldwide. For its specific characteristics, the battery qualified to the finals of the NASA iTech competition of technologies for space missions. It also prides itself over its robustness, nonflammability and recyclability.

In the early stages of the battery development in 2016, the battery was supported by a single investor acting as a business partner of the inventor. However, with the continued development of the technology, further investment was needed. The technology found investment from a German-based arm of a Chinese investment company. This strategic partnership however ended swiftly in a lawsuit, the investor claiming a breach of the contract, HE3DA accusing the investor of industrial espionage and an attempt at stealing the trade secrets behind the battery.

Despite this troubled start, HE3DA project managed to take off. With the help of Battery Unite, a local entity established to gather investments for HE3DA from private sources (including businesses, industry and the financial sector), HE3DA opened its first factory in 2020. Starting first with a pilot production, the serial production should commence in 2021 - first at the capacity 200MWh/year but reaching 1.2 GWh/year by the end of 2021. The aspiration is to keep enlarging the production eventually up to 15 GWh/year.

According to the information from HE3DA, the fundamental research has cost EUR 10 million, the development of the technology EUR 2 million a year and the construction and technology of the factory amounted to approximately EUR 52 million. These financial sums were leveraged from interested private investors. HE3DA demonstrates that while not being an easy exercise, the commercialisation of battery technology is possible even without public funding.

The second important aspect for bringing novel battery technologies to the market is the existence of infrastructure for piloting and demonstration of the battery innovation in question. Demonstration projects and pilots are important to test out new technologies in near-market conditions, prior to ramping up the production on a commercial scale.

Europe has already several pilot lines for Li-ion batteries in operation (*e.g.* at ZSW, CEA, Fraunhofer IKTS, IK4 CIDETEC), and new lines are under construction (*e.g.* TNO's pilot line for new 3D solid-state lithium-ion batteries). They are key facilities to help in scaling-up the electrode and cell technologies by developing a prototype that proves whether the technology works and can be mass-produced. A new project under Horizon 2020 called LiPLANET111 will map the existing and forthcoming EU lithium battery cells pilot lines to analyse the potential gaps and synergies among them. It will eventually bring them together in a dedicated network to facilitate knowledge sharing and access rights between these pilot lines.

Europe is also home to several demonstration plants. Demonstration plants can generate significant quantities of the product for market testing and proving its commercial and technical viability. Most demonstration plants focus on testing the Li-ion cell technologies, there are however also pilot plants for testing innovation for battery end-of-life processing.

In detail: Fraunhofer Research Factory for Battery Cells[112]

To promote the battery innovation and commercialisation of Li-ion battery technologies, Fraunhofer partnered with universities in Münster and Aachen in a demo battery factory project. The demo research factory will serve as a testing facility focused on battery technologies with advanced technological maturity (TRL 6+).

These mature technologies will be tested for their proof of reproducibility and scalability at the industrial scale providing the investors and industry the assurance of the technology's aptitude for mass production.

Nevertheless, it should be noted that a number of these pilot and demonstration plants are in their early stages and are thus not yet in full operation. In addition, there is a lack of specialised demonstration projects – for example in systematically gathering data about the added value of battery integration in the energy system as an alternative to conventional grid reinforcement.

The barriers to technology transfer and commercialisation of battery technologies

Many significant improvements have been made in recent years in technology transfer and the commercialisation of battery innovation. However, one of the main points of criticism is that the current speed of commercialisation is still insufficient to respond to growing market demand and become competitive in the global market. Momentum will have to be kept in the years to come and the speed of maturation will have to be increased for upcoming battery innovation.

Fierce competition has proved to be one of the facilitators of technology transfer and the commercialisation of battery technology. However, the competition also makes investment into batteries a riskier business. Indeed, bankruptcies and takeovers of battery companies have not been (and are not) exceptional. Moreover, competition may also act as a barrier to the market uptake of novel technologies as they will have to compete with the existing generation of Li-ion batteries. In order to speed up the construction of battery gigafactories in Europe, it is necessary to focus on the deployment of already-existing technologies. However, to avoid stifling the other breakthrough technology developments and losing the future technological advantage, sufficient support also needs to go to the development of next generation batteries.

Public intervention plays a crucial role in making the business case for batteries and in supporting the commercialisation of battery technology throughout its development. Improvements of the current system are nevertheless still needed. The European infrastructure for piloting batteries should be finalised and the network of these facilities should be coordinated to avoid duplication and to ensure targeted specialisation where necessary.

As established players may be hesitant to commit to unproven technologies, it is start-ups that are often at the forefront of the commercialisation of breakthrough technologies. This is especially the case in smaller niche markets or in energy storage solutions.[113] The associated risks may however deter private investors, so public funding needs to play a more significant role in supporting start-ups in their efforts to scale up promising early-stage technology and bring it to the market.

There are several public support instruments for financing the commercialisation of battery technologies. However, as the ambition to develop the market is high, the need for ramping up investment is still greater. While Europe directs public funding primarily into research, its Asian competitors systematically fund the deployment of batteries and battery manufacturing infrastructure. Therefore, the need for amassing finance for commercialisation and scale up is unceasing. Bearing in mind the long timeframes for a battery technology to go through the technology maturation process and reach the market (over a decade), public instruments should provide long-term financing opportunities that prevent the premature sale of the technology due to the need to recoup the investment.

A coherent strategy and regulation also play a crucial role in shaping the European battery sector. The EU's quest to produce batteries that are sustainable is part of its competitive strategy to position itself in the already dense market. An EU-level regulation setting environmental standards for batteries produced in the EU is generally welcomed by stakeholders, as long as it provides clear legislative framework and does not place the European production at a competitive disadvantage as compared to global competitors.

Moreover, stakeholders argued that overly detailed standardisation in the field of batteries where the optimisation process is not finished could potentially stifle the development of alternative battery technologies that do not correspond to the standard adopted.

With regard to the battery value chain, the need to ensure self-sufficiency in the provision of raw materials persists. Public support, frontloaded financing and fast permitting could well assist in the development of raw material mining and processing in Europe. On the other side of the coin, while the deficient supply of raw materials may be an incentive for innovation (developing battery chemistries without certain minerals), it may also represent a barrier to successfully moving battery cell production to Europe in general.

In terms of grid-scale energy storage, several barriers to the commercialisation of the innovation for energy storage have been identified. Energy storage applications have been overshadowed by the rapid developments on the EV market. Li-ion batteries are therefore currently the most used technology both for EVs and for energy storage. Yet the battery characteristics for grid energy storage differ from those of EVs. Moreover, the energy storage market has specific needs for public support for deployment projects, funding and enabling regulatory framework in order to take off. Some of the identified barriers specific to energy storage were lack of investment and business incentives (including feed-in tariffs, price signals and time-off tariffs, long-term contracts, public funding or public procurement or demonstration projects specialised in testing the battery integration in the grid), overlapping legislation, double charging of energy storage devices and the lack of well-functioning markets. These general problems may also disincentivise the commercialisation of battery innovations for energy storage.

Finally, the general lack of skilled human capital both in battery research and (especially) in applied process design has resulted in 70% of companies delaying their investment decisions. Unless a skilled workforce is attracted from abroad or (re-)trained, this problem will only grow with time.

Conclusions: Case study on battery technology

Despite some of the challenges and barriers to technology transfer and commercialisation of battery technologies in Europe, there seems to be much optimism for the future. The EU has made its intention clear to bring battery production to Europe. This, coupled with an emerging regulatory, funding and coordination framework, has helped to activate stakeholders along the entire value chain. It represents first steps towards establishing a unique European battery ecosystem.

While most of the focus is on creating mass production of already mature battery technologies and building Europe´s first gigafactories, other battery technologies also benefit from this upswing in public attention to the battery industry. Thanks to intensive coordination activities at the EU-level, public research is generally well coordinated along current or future market needs. This helps to avoid duplication in research activities and enables smoother technology transfer of innovation from universities to the market – whether to existing companies or to start-ups.

It is self-explanatory that much remains to be improved. The previous section shed light on some of the existing barriers to and needs for streamlining of technology transfer and commercialisation processes in the battery field.

While it remains yet to be seen whether Europe manages to become competitive in the global race in battery production, the public policy focus and funding has already achieved remarkable results. In this regard, many of the following lessons from the battery sector may also be applicable to other green technologies.

First, significant efficiencies can be achieved through intensive coordination among R&I and the industry – whether in the form of sector-specific alliances and platforms or joint research projects. This collaborative ecosystem also encourages knowledge transfer.

Second, while rapid diffusion of already mature technologies and ramping up the manufacturing in Europe may be key to ensuring competitiveness in the battery cell production, the focus on deploying mature technologies should not result in rewarding only the current “technology winners”. Indeed, enabling environment should be strengthened to allow other novel technologies to mature and eventually replace the current ones. The regulatory framework and technical standards should not be narrow to the extent that would prevent future development of alternative solutions.

Third, innovative technologies need sufficient public support, especially in amassing sufficient finances for the scale up of the technology. In particular, this applies to start-ups that were created around new early-stage technologies. Public support – whether in the form of funding, investment, guarantees or other – is typically key to attracting private investment for a particular venture.

Finally, the integration of the whole value chain, from the provision of raw materials to the manufacturing of batteries and end-of-life disposal is key to ensuring self-sufficiency, a circular economy, and more sustainable batteries in future.

Altogether, the interviewed battery stakeholders in principle agreed that Europe is on the right path towards shaping a new battery industry and that EU policies promote the commercialisation of novel battery technologies. It is important to keep this momentum.

5.3 Case study: Artificial Intelligence

In short:

- **Artificial intelligence** is a general-purpose technology that has a strong potential to make a wide variety of processes or sectors more sustainable. On the opposite end, as AI is poised to become a mainstream technology, “greening” AI, such as developing more efficient AI algorithms, greening data, and greening AI hardware and infrastructure, will play an important role to limit environmental footprint of AI. The lack of a robust regulatory framework was brought up by stakeholders as posing a barrier to the development and commercial use of AI in the EU. Moreover, European stakeholder networks should be further facilitated to achieve stronger coordinated approaches to the use of AI and to incentivise greener AI innovation. Stakeholders highlight EU strategies such as the Digital Europe Programme, Horizon Europe, and regulatory sandboxes as potential facilitators for AI technology transfer and commercialisation.

Seventy years ago, Alan Turning asked the question whether machines *can think* and perhaps on *one day learn*.[114] In modern society, artificial intelligence (AI) is no longer an idea on paper; it is permeating into everyday life, such as in the areas of mobility, manufacturing, advertising, health care, agriculture, entertainment, government, and education. Some scholars and policymakers refer to AI as having the potential to become the next major milestone in human development. Some say that AI will reshape our environment, interactions, and daily practices.[115] Whereas others are lauding that, a *revolution* is underway, where AI or digital brainpower is comparable to the magnitude of impact that mechanical power and computing power had on societies during previous revolutions.[116]

There are high expectations that the functionalities of AI will continue to grow over the years to come.[117] Some are of the opinion that AI will become smarter than people are in the near future.[118] What this would mean to life as we know is subject to fierce debate; some are believe that AI could pose a significant threat to democracy.[119] For now, it is still to be seen whether any of these expectations will come to fruition. Certainly, the policy choices that we make today will have a lasting impact on the way in which we will live with AI in the future.

What is AI precisely?

The definition of AI is subject to strong debate[120]. Rather than viewing AI as a single technology, a common viewpoint across academic literature is that it refers to a range of sub-disciplines and techniques that can undertake tasks with a degree of autonomy.[121] To understand AI well, we will briefly discuss AI's core components below.

Algorithms are one of the key components that permit AI to function. Algorithms carry out a sequence of steps to complete a task, these steps must be described accurately (in a programming language) for a computer to run it.[122] AI algorithms are often complex,[123] and in some instances to such an extent that even AI experts are unable to unravel AI algorithms.[124]

Big data can be classified across the four factors of [125] volume, velocity, variety, and complexity.[126] In a general sense, data can be referred to as information, and big data underlining the volume of information that is involved.[127] In this report, data or big data refers to the information that AI algorithms require to carry out their tasks.

Computing power is essential to AI. In general, the robustness of AI has an impact on the amount of computing power that AI needs to function. The increasing affordability of computing power has been a strong enabler of the rise of AI in modern day society. Improvements of energy efficiency of computing power has substantively contributed to the affordability of computing power; over the past years, the number of computations per unit of energy has doubled approximately every 1.57 years.[128]

AI has **intelligent, automated, and cognitive capacities.** Automated decision-making refers to the ability to take a decision without the need for a human to intervene.[129] Cognitive capacities can refer to functions such as recognition, thinking and learning[130].

In AI, the level of intelligence is often expressed as **strong AI or artificial general intelligence (AGI)** versus **weak or narrow AI.** Weak AI generally refers to AI that is specifically designed to execute singular or specific tasks.[131] The AI applications that exist in present day fall in the weak AI category. Strong AI is expected to surface in over a decade; it refers to AI that has the ability to execute true automated and cognitive functions, including being self-aware.[132 133]

One of the prominent fields of AI is the domain of **machine learning**. Machine learning refers to models where algorithms learn from the data that they process.[134] **Unsupervised learning** models can uncover patterns in data that is unclassified. **Supervised learning** models are able to make predictions based on classified data, and is generally used to train a model to make accurate predictions.[135] Another sub-category which is subject to a high degree of attention is **deep learning**, which is an algorithmic model that can autonomously extract features, make predictions, and identify patterns from unclassified data.[136] The fundamental structure of a deep learning model is something that is called a **deep neural network** – a reference used because the model is inspired by the functioning of the neutral network in the brain.[137]

EU policy approach

Milestones that strengthened the focus on AI at the EU policy level are the Declaration of Cooperation on Artificial Intelligence presented in April 2018 together with, the Artificial Intelligence Strategy, and the Coordinated Plan on AI (developed together with Member States) of December 2018. In May 2019, AI was mentioned in the European Council's Leaders' Agenda, and thereafter it was placed under one of the four priorities of the EU's Strategic Agenda for 2019 - 2024. In July 2019, Ursula von der Leyen, President of the European Commission, outlined in her Political Guidelines the aim *"to put forward legislation for a coordinated European approach on the human and ethical implications of Artificial Intelligence", — Ursula von der Leyen* adding that investments in AI are to be prioritised through the Multiannual Financial Framework and public-private partnerships.[138]

While AI falls directly under the priority a Europe fit for the digital age, its potential to contribute to the EU's green agenda is clearly referenced in the European Green Deal. The European Green Deal communication writes that *"digital technologies are a critical enabler for attaining the sustainability goals of the Green deal in many different sectors. The Commission will explore measures to ensure that digital technologies such as artificial intelligence… can accelerate and maximise the impact of policies to deal with climate change and protect the environment".*[139]

The vision of harnessing the potential of digitalisation to achieve the EU's sustainability goals is also reflected in the EU's digital strategy. Specifically, the communication on shaping Europe's digital future makes a reference to the planet as being one of the beneficiaries of the strategy; under its third pillar an open, democratic and sustainable society, the communication writes that the aim is to *"use technology to help Europe become climate-neutral by 2050" and "reduc[ing] the digital sector's carbon emissions".*[140]

In February 2020 the Commission presented a white paper on artificial intelligence – a European approach to intelligence and trust. Already in the first paragraph, the white paper writes that *"AI… will change our lives by… contributing to climate change mitigation and adaptation. Moreover, in the introduction, the paper mentions that "digital technologies such as AI are a critical enabler for attaining the goals of the Green Deal"… [and that] the environmental impact of AI systems needs to be duly considered throughout their lifecycle and across the entire supply chain, e.g. as regards resource usage for the training of algorithms and the storage of data".*[141] The part referring to AI as an enabler of the Green Deal concerns applying AI as mentioned in the box below, the second part concerns green(er) AI.

How does AI relate to sustainability?

AI is a general purpose technology, as in, it can virtually be applied to any field. This is why AI is applied in sectors that may be unrelated to each other. There are two areas where sustainability counts most when it comes to AI, these are: green(er) AI, and AI applications for sustainability purposes.

Green(er) AI, is a field that specifically looks at reducing the environmental footprint of AI. One example is improving algorithms in a manner where desired functions are carried by using less computing power, thereby consuming less energy.[142] Another example is reducing the footprint of hardware and infrastructure components used to run AI. Overall, green AI focuses on greening the core components that AI needs to operate.

The second area where AI can have an impact on sustainability is by integrating functions in **AI applications** that can reduce the footprint of certain processes or sectors. One example would applying AI in the field of mobility, such an AI-improved traffic light system, which can help to decrease the amount of traffic and significantly reduce carbon emissions.[143] Another example is in agriculture, where an AI system specifically learns how much fertilizer to apply to crops throughout the year, or how to recognise ripe produce, thereby reducing the environmental footprint of growing and harvesting crops.[144]

Beyond the EU's policy framework, a variety of AI networks exist at the European level where stakeholders can engage. The Confederation of Laboratories for Artificial Intelligence Research in Europe (CLAIRE) consists of a research network to which 385 labs and institutions are linked, and an upcoming innovation network. Coordinated by Aalto University and partially funded under Horizon 2020, the European Network of AI Excellence Centres (ELISE) is a network of AI research hubs. ELISE is building and executing a European Strategic Research Agenda in AI, it has a PhD and Postdoc programme, contains a researcher mobility programme, and it fosters industry collaboration. The European Laboratory for Learning and Intelligent Systems (ELLIS) brings together Europe's top researchers who are active in AI, they also have a fellowship programme that appoints fellows and scholars.

The European AI Alliance is a multi-stakeholder forum with over 4000 members. It was launched in 2018 following the EU's AI Strategy. The AI Alliance provides a forum where daily exchanges take place, feeding into the European debate on AI and the European Commission's policy-making. The European Association for Artificial Intelligence (EurAi), established in 1982, aims to promote the study, research, and application of AI in Europe. EurAI holds annual conferences, organises courses, writes out awards and grants, and sponsors conferences and European AI educational programmes. The Big Data Value Association (BDVA) and the European Robotics Association (euRobotics) are two public-private partnerships under Horizon 2020. The European Partnership on Artificial Intelligence, Data and Robotics will build on the work of BDVA and euRobotics as a public-private partnership under Horizon Europe.

Technology transfer and commercialisation of AI technology

While the AI white paper already offers some policy direction for AI stakeholders, legislative clarity is still pending. At present the European Commission stopped short in early 2020 of presenting legislation on AI, a legislative proposal on AI is now anticipated in early 2021. The absence of clear rules on AI in the EU can pose challenges for current AI innovations in the EU. For example, current AI innovations may fall short of the criteria in the upcoming legislative proposal. A concern is whether low-risk AI innovations will be subject to the same legislation as high-risk AI innovations. There is also a concern whether all sectors will face the same rules, as the rules to apply may be context dependent. There are also concerns about value creation and taxation, especially in light of the potential risk of tax avoidance by big players, making it challenging for smaller AI innovators to compete. Finally, there are concerns surrounding interoperability of AI technology and policy fragmentation, that may stem, among other reasons, from differences in the legislative environment on AI per Member State.

Data was found to be a key facilitator and barrier to AI innovation, depending on the situation. Challenges exist around collecting data, storing data, the type of data (personal data versus machine data), processing data, labelling data, sharing data, filtering data, quality data, AI training data, and access to data. To some extent innovators face constraints under the current GDPR regulation; and despite the GDPR, data privacy concerns continue to permeate. Another challenge regards the monopolistic risks of the data market, where only big players control the necessary data to train AI and bring new AI applications to the market, thereby excluding smaller players.

As mentioned in the AI white paper and the EU's digital strategy, it is important to ensure that we are reducing the footprint of AI technologies and digitalisation in general. At present, a barrier exists for the development of green(er) AI in the EU. There is a limited amount of funding available under the current EU's financing instruments instruments in greening core AI components (algorithms, software, data), hardware needs (chips, other computing components, and robotics), and infrastructure (connectivity of AI technology, databases).

Moreover, the deployment of AI technology in the EU can be further facilitated with real-world testing and experimentation facilities. The absence of such facilities created challenges for example under Horizon 2020, where pilot projects did not go through real-world testing, and encountered challenges in the technology deployment phase.

Setting up a strong network on AI, where key stakeholders can interact on a regular basis and provide input to AI policy, was found to be a key facilitator for AI innovation in the EU. It was found that, as AI is a general purpose technology, industry and research teams should be firmly embedded in these networks.

Concerns were found regarding the EU's competitiveness on AI compared to the US and China. In the EU, a low amount of funding and investment is available for the development of AI innovations compared to the United States and China. Questions surface whether this would have an adverse impact on the EU's ability to compete with AI innovations from the US and China in the medium and long-term.

Another finding on the competitiveness of the EU regards is the amount of computing power that is necessary to train AI models.[145] Over time, it is anticipated that only a few large players will have the necessary super-computing infrastructure needed to develop next generation AI innovations. Of special concern is whether this computing power can be delivered in the EU by EU players, or whether it will be tech giants in the United States and China.

Many AI developments are currently taking place in an open source environment. The facilitating effect of open source AI is that it permits access to a large AI developers pool, which is beneficial to overall innovative progress. Question marks remain whether open source development poses a barrier to the ability to commercialise AI innovations. Licensing could act as a facilitator when it acts as a means to reward the innovator. Potential barriers of licensing are reduced flexibility due to strict licensing conditions, and expensive licensing fees. Another challenge regards copyrights and the ownership of output that is created by AI[146]. Questions arise whether the owner of the output would be the AI developer, the AI handler, or the AI application itself. Further inquiry is still needed to discern whether patenting[147] and trademarks act as facilitators or barriers to AI.[148] The overarching issue of IP policy and AI is now subject to discussion at WIPO.[149]

Conclusions: Case study on artificial intelligence

The upcoming legislative proposal on AI is highly anticipated and needed to provide necessary clarity on the course of direction for AI in the EU. While it is essential to keep the EU's value-based and human-centric approach to AI in focus, it is recommended to strike the right balance to account for heterogeneity of AI applications. In that regard, it is recommended to consider how rules would apply depending on the context, sector, and risk-level of the AI application. Moreover, it is recommended to address the issue of value creation and taxation, especially to prevent big players from gaining a competitive advantage over smaller AI innovators. Finally, issues of interoperability of AI and policy fragmentation should be addressed to ensure that new AI innovations can be rapidly deployed across all Member States.

It is recommended to ensure that the EU's approach to data in the upcoming legislative initiatives, such as the initiative on Common European Data Spaces, the Data Act, the Review of the Database Directive, aim to facilitate data needs for AI innovation, as identified in the section above. For green AI, a special area of importance is data efficiency, by ensuring a strong approach towards greening data processing, and by finding routes to further encourage research and technology organisations to carry out more research on data efficiency.

It is recommended that more funding becomes available for innovations and projects that focus on the development and the deployment of green(er) AI. There are currently no dedicated calls for proposals that focus on green AI. Key venues to address this are through Horizon Europe and the Digital Europe Programme, but it is also recommended to ensure funding through other funding instruments, for example funding for green AI infrastructure needs through the Connecting Europe Facility. An additional venue that could be considered is setting up a dedicated research programme on green AI.

To this end, it is also of recommended for the EU and Member States to further incentivise green(er) AI innovation. Incentives can trigger a market demand for more green(er) AI innovation, which can be a strong motivator for research and technology organisations to ramp up their activity in this space. One option would be a label for green(er) AI, based on a model that inspired by other sustainability labels, such as labels for eco-products. It is also recommended to consider placing energy efficiency standards for datacentres, as there will be a rapid growth of datacentres across the EU in the decade to come that will in part power the data needs of AI applications. Finally, it is recommended to consider green procurement, for example by letting procurement to be dictated by the greenness of an algorithm or other core AI components as selection criteria.

Embedding a clear approach in the Digital Europe Programme on establishing a real-world testing and experimentation environment can facilitate to ensure a better deployment of new green innovations. In that regard, establishing a regulatory sandbox can be of additional benefit to allow for risk management, allow for dynamic learning by innovators and regulators, and to allow innovations to reach the market faster.

The EU's AI competitiveness should be addressed as early as possible. To bring investment in AI to a comparable level as in the US and China, it is recommended to continue strengthening the EU's funding mechanisms that target private investors. Examples of such mechanisms are the EU's Artificial Intelligence and Blockchain Investment Fund and the EIB's Artificial Intelligence Co-Investment Facility. Moreover, it is recommended to plan for the EU's computing power needs early on - to ensure that the appropriate computing infrastructure (*e.g.* super computers) are available and affordable within the EU, and provided by EU players. A parallel path to halt excessive computational power needs is through increasing computational-efficiency and energy efficiency of algorithms under the banner of green(er) AI.

It is recommended that the upcoming updated coordinated plan on AI with Member States will offer a perspective on coordinating on AI for sustainability and green(er) AI, including concrete venues for cooperation on innovation and technology deployment.

The EU's and European stakeholder networks on AI should be further strengthened, and it should especially be ensured that industry and research teams are firmly embedded in these networks. In this regard, it is recommended to organise events and conferences specifically dedicated towards green AI and AI applications that deliver on achieving the EU's climate and environmental objectives.

5.4 Case study: Carbon Capture, Utilisation and Storage

In short:

- The case study on carbon capture utilisation and storage shows that technology transfer in the field does not constitute a major challenge as industry players are the main driver of the innovation in this field. Commercialisation, however, faces numerous barriers. Public support is needed to significantly decrease the costs borne by first-movers and to de-risk investment and initial operation over the time horizon and across the value chain. Only a few demonstration projects exist and further processing or end-users are missing, which leaves investors without a clear business case. Public acceptance and political commitment must be strengthened further in order to create certainty for investors, whose return will materialise over a longer timeframe. The regulatory framework still needs to be improved to enable cross-border operation for the industry, which would benefit from operating as part of a broader European network.

Carbon capture is a promising solution to mitigate the impact of CO_2 on the climate. Carbon capture and utilisation (CCU) is the technology to separate CO_2 from emitted gases of industrial processes and reuse it directly or as a feeding stock for other industrial purposes. In the case of carbon capture and storage (CCS) the final aim of the capture is to transport the CO_2 gases and inject them into deep underground rock formations. These technologies have the potential to contribute to reach climate neutrality, to create a circular economy and industry and to reduce the emission of greenhouse gases as envisaged in the European Green Deal[150]. The EU aims to become a frontrunner of breakthrough technologies and the Green Deal ambitiously aims to commercialise CCU and CCS in key sectors by 2030.

CO_2 captured through CCU technology could be used among others for the production of dry ice, pharmaceutical and construction materials and could play a role in energy storage and fuel production. Hence, there is a wide range of industries, which could uptake the technology once it reaches the commercial scale and competitive price levels. This includes the aforementioned production of *"blue" hydrogen.*

Nevertheless, its recognition as a climate mitigation tool is debatable if its high-energy usage does not come from renewables and if the captured GHG gets still released to the air at a later point in the reused gas' lifecycle. CCS provides a solution against emitting CO_2 gases in the air, but its environmental impact must be carefully considered, the transportation and the storage require thorough precaution and it raises legal questions in the field of waste disposal and liability for damages.

There is a great climate mitigation potential in carbon capture, utilisation and storage technologies, but for the moment there are numerous financial, political, sustainability and legal challenges that can hinder the commercialisation and the industrial use of these tools. Therefore, it is particularly relevant to examine the deployment of the technology in a case study in order to explore the barriers to the transfer and commercialisation of these technology, which could shed a light on commonalities with other green technologies.

EU policy approach

The European Green Deal has highlighted CCUS technology as an innovative solution with high potential to contribute to achieving net zero emission by 2050. This target has also been laid down in the European Climate Law, which will set in law the reduction of greenhouse gas emission by at least 55% by 2030 and to net zero by 2050[151]. Furthermore, the CCS Directive, the Taxonomy Regulation and the Emission Trading System established a clear regulatory environment CCUS projects, although there is room for further improvement as it will be discussed later in this case study.

The European Strategic Energy Technology Plan (SET Plan) aims to foster the green transition and the development of low-carbon technologies by "bringing down costs through coordinated national research efforts. The SET Plan helps promote cooperation among EU countries, companies and research institutions, and in so doing also deliver on the key objectives of the energy union.[152] Within this frame the Implementation Working Group 9 coordinates and promotes the research and innovation activities for CCUS. The updated Implementation Plan lays out 10 concrete targets to be achieved by 2030 in delivering commercial scale deployment of carbon capture projects.[153] On the level of individual Member States, the National Energy and Climate Plans were adopted in 2019, several of which outlines interest and future research activates in CCUS.

Lastly, CCUS research and demonstration activities are also linked to the Trans-European Networks for Energy (TEN-E), which is a policy that is focused on linking the energy infrastructure of EU countries.

This policy helps to connect energy infrastructures across Member States, which is particularly relevant for the transportation of CO_2 and for the creation of the CCUS value chain.[154] Moreover, Horizon 2020 and Connected Europe Facility (CEF) also provide funding to CCUS projects.

Technology transfer and commercialisation of CCUS

While the research in the CCUS industry is ongoing, nevertheless, the transfer of technology is not being held back and it is not considered as a major point of concern by stakeholders. CCUS technologies has been available on the market for several years and they are considered mature enough for industrial use. Research and the transfer of new technologies are necessary for improving efficiency or developing new industry use cases of the technology. Therefore, research activities are mainly driven by the industrial demand side and are primarily conducted by businesses active in the field, while universities tend to make less patent applications related to CCUS than industry.[155] As a result, technology transfer often happens within the industry, which makes the transfer to businesses easier, but makes the innovation driven by market pull effect.

First generation capture technologies have sufficient level of maturity for market deployment and commercial use. While research continues to explore the optimisation of these technologies but research focusing on second and third generation technology solutions is also already in progress. The main technology methods for capturing CO2 is summarised in the table below.

SOLVENT-BASED CAPTURE	CO2 is being absorbed chemically or physically into a liquid carrier
SORBENT-BASED CAPTURE	CO_2 gets separated through pressure or temperature swing absorption process
MEMBRANES	Membranes use (semi-) permeable materials to separate O_2 from flue gas
HIGH TEMPERATURE LOOPING SYSTEM	O_2 separation in boilers using O_2 carriers that oxidise at high temperature

Table 2: Main methods of carbon capture[156]

The efficiency of capture methods can be improved or cheaper methods and new use cases can be developed, however, their deployment would face similar difficulties if the deployment is hindered by more systemic barriers to commercialisation. At this point, it seems that cost reduction achieved through enhanced efficiency or new use cases are not in the limelight of the policy debate since discussions signalled the obstacles to commercialisation as most prominent barrier to deployment. Therefore, this case study focuses on the economic, political and regulatory barriers to commercialisation as main categories identified by the interviewed stakeholders as most prominent barrier to deployment. Therefore, this case study focuses on the economic, political and regulatory barriers to commercialisation as main categories identified by the interviewed stakeholders.

Economic barriers to commercialisation of CCUS

Probably the most important barrier above all is: price. It is well-known that any green technology can only be sustainably commercialised if it is worth to be implemented by business, meaning that it generates enough profit and it is less costly or more reliable than the existing alternatives. This is also the case for CCUS, since market players still find emitting more profitable than capturing the CO_2, and this technology has to also compete with other solutions in certain sectors such as renewable energy production. Therefore, the business case for CCUS needs to prove that such technology can provide viable and profitable alternative revenue streams compared to other solutions.

This will not be achieved through reaching higher efficiency in the near future. Stakeholders from the private sphere were aligned on not seeing high-cost reduction potential in achieving higher capturing efficiency. Instead, the preferred way forward is making polluting more expensive and by that directing the industry towards the increased use of CCUS technologies, which would drive down the cost due to the economies of scale. The carbon prices of the ETS are generally considered as too low, which do not provide enough incentive to implement other greener solutions. ETS is a complex system and carbon prices are not the sole yet an important factor for deploying alternative solutions such as CCUS. The revision of the scheme is well in progress, which can give an important push to commercialisation. Nevertheless, more expensive emission does not constitute a business case on its own, and it does not make CCUS competitive against other sustainable solutions such as renewable energy sources. Therefore, some stakeholders argued that the focus of the discussion should be moved towards demonstration projects creating the business case instead of arguing over ETS prices, which would crowd in investment in the technology. This could reduce the costs through the economies of scale, hence making CCUS more competitive. But reaching such level of competitiveness requires access to early financing, which is troublesome for CCUS projects for numerous reasons:

Banks and venture capital are not willing to take the risk to invest in these projects. Running CCUS projects often search financing from other companies and public bodies investing in their project, because there is no business case to be presented to the financial sector. Actors in the financial sectors need to understand the risk attached to these projects. Most significantly, the risk related to the time horizon and to third-parties of the projects are prominent barriers to attracting investments. Regarding the third-party risk, there can be multiple players involved in the operation of CCUS value chains including parties who capture, those who operate pipelines and transport infrastructure and those who reuse or store the CO_2. The involvement of multiple parties makes it difficult for investors to assess whether investment in a CCUS facility will be backed up by sufficient transportation infrastructure and capacities and whether the demand from the market and storage capacities for captured CO_2 will be there in the long run. The time horizon risk is closely related to this, since it is uncertain how the market demand for capturing and the political landscape (see next section) will evolve over the investment horizon. That makes return prospects uncertain for investors, therefore it is necessary for policy makers to de-risk CCUS related investments by providing public guarantees over the value chain including providing long-term assurances for storage sights.

With regards to financial instruments, policy makers have a key role in making the business case operational, since the cost of first movers is always higher and that is valid for most innovative technologies. Nevertheless, the composition of financing instruments can be crucial. The EU has made funds available for demonstration projects from Horizon 2020, Innovation Fund and for projects of common interest from CEF. Although there can be always more funds made available for the demonstration of these technologies, but future public financing should better coordinate the efforts of different regional/ national/ EU funding streams, which should also go beyond financing the CAPEX related expenses of the demonstration and finance or de-risk OPEX expenses of first movers. The support of certain operational costs should be also paired with enabling investment across the value chain in order to create a functioning ecosystem for CO_2 capturing, transport and storage/ utilisation. These financing measures then should be replaced by guarantees and be phased out gradually in order create the business case, which could attract investors and crowd-in private capital. These findings were not just confirmed by the industry but also by actors of the financial market and can be potentially true for other green technologies.

Political barriers to commercialisation of CCUS

As it has already been mentioned, political barriers have accompanied the history of CCUS implementation in Europe. The early 2000's brought an increased interest in the technology, which was later put on hold as policy makers were looking elsewhere and the lack of business case remained a barrier to investment. Now, as CCUS has become part of the EU's Green Deal, there is a momentum to step up the support of CCUS deployment. The bottom line is that green technologies are changing the way people used to approach certain things (such as consumption patterns) and they transform our societies. Therefore, it is vital for successful and long-lasting deployment of green technologies to be well-perceived and embedded in the society. Public acceptance can majorly influence the success of projects and the choice of policy makers in many fields from wind turbine farms to COVID-19 vaccines. Public acceptance of CCUS has not been very high in the past and it may be influenced by several factors such as planned proximity of capturing or storage facilities. However, as the urgency of reaching net zero emission increases the potential of this technology is being recognised, although the mayor concerns due to *"end-pipe-solution"* nature of the technology and due to maintaining CO_2 production remain.[157]

If policy makers decide that CCUS is a necessary tool for decarbonisation then there should be a clear sign of commitment in order to eliminate political uncertainty for investors. As the time horizon for such investment exceeds the length of political cycles, therefore, there is a need for political consensus on the matter and it should be incorporated as part of a long-term decarbonisation strategy. This could assure investors that EU Member States are counting on this technology on the long run, therefore, the political risk associated with the long time-horizon of the investment could be reduced. Here the multiannual financial framework (MFF) of the EU and the European Commission could play an important role as a facilitator, since the lifetime of the MFF and the Commission's role to monitor the implementation of EU wide strategies exceeds the short political cycles of national governments. Although determining the pathway of greenhouse gas emission reduction is up to the Member States, as it has been laid down in the national energy and climate plans[158], but the EU can coordinate and facilitate the implementation of these plans and promote a role for CCUS, if such policy choice is made.

Furthermore, the public could take an active role in de-risking certain activities of the process. In particular the risks linked to storage sites is an important barrier in this regard. There is no clearly definable timeframe attached to storage activity, as it is practically limitless in time, therefore, the associated risks and total costs remain unknown for private investors. It should be clearly defined who will take the costs for storage in twenty or fifty years and who will take responsibility for any malfunctioning or leakages in the distant future. Policy makers could provide certainty if they limited the time horizon by giving the state a guarantor role or by taking over the storage sites after defined years of operation. Hence, the provision of a policy framework with appropriate incentives that enhances the predictability of long-term investment plays an important role in the creation of a business case. This can be equally valid for commercialisation of other green technologies.

Regulatory barriers to commercialisation of CCUS

The EU has a solid regulatory framework for CCUS and for CO_2 emission, which is mainly driven by the Directive on geological storage of carbon dioxide, by the Taxonomy Regulation and the ETS rules. The legal framework provides a rather clear situation for operating CCUS and storing CO_2, however, there are some key issues that are still open and should be addressed in order to accelerate and support the deployment of CCUS technologies.

The most frequently citied regulatory barrier is related to the transport of CO_2 within the EU. A viable CCUS sector requires the cross-border flow of CO_2 in the Single Market, since the nearest using or storage facility may be in another Member State, and this is particularly important in case of sea and waterway transportation. The cross-border movement of captured CO_2 is regulated in the London Protocol, which lays down the foundation of international CO_2 transport. In the EU increasing number of Member States have signed the Protocol, nevertheless, there are still a few EU countries, who should sign or ratify the agreement.

Once the carbon-dioxide can legally be transported between countries then it becomes necessary to have certain rules for quality in place in order to ensure that the CO2 that is put in the pipelines or that is being moved from one country to another qualifies to the same quality criteria on all sides. EU level guidelines could set the minimum requirements. Having such guidance in place early on would support the formation of the market in this early stage, it could ensure compatibility across the EU and could prevent difficulties coming with large scale deployment.

Another aspect that would require legal clarification already for early demonstration projects is the legal liability of storage sites.

There is indeed a rich literature of this problem discussing, whether in case of any damages or leakages the liability should lie with the emitter of the CO_2, with the operator of the storage facility or with the state. This legal uncertainty can hold back companies and investors from engaging in storage activity. It also makes cross-border transportation of CO_2 more complicated, as it is unclear who should bear the responsibility for stored gases coming from another Member State, and this could create frictions between Member States in the future. Due to its cross-border nature, it is necessary to clarify these issues at the EU level. This would support good cooperation for joint CCS projects of Member States and reduce the risk for private investors by providing legal certainty.

Lastly, the provisions of the Taxonomy Regulation and the ETS affect the deployment of CCUS and in particular CCU negatively. It is still being debated whether CCU mitigates the CO_2 emission effectively, and therefore, it has not been included as an eligible technology under the Taxonomy Regulation, nor any CO_2 transport assets related to CCU. The issue is being discussed in Taxonomy Platform, but should the EU decide to change its policy approach that would require also the revision of the ETS. Besides the fact, that the cost of emission is generally considered to be too low, which was confirmed by the consultation with stakeholders, a revised ETS should make sure that it all modes of CO_2 transport – *e.g.* pipeline, ship, barge, train, truck – are approved in line with the rules of the Taxonomy Regulation. If CCU is to be supported, then ETS should also enable CCU by not having to give up allowances if the utilisation leads effective storage of the CO_2, like for example in the form of mineralised construction materials. Indeed, in light of the ambitions of the Green Deal both the Taxonomy Regulation and the ETS are currently under revision. Putting aside the policy debate about the effectiveness of CCU as an emission mitigating tool – which is not subject of this report – clear policy choices will have to be made in order to send a strong signal and provide certainty for the industry and investors in the long-term. Furthermore, easy transportation of CO_2 stream with harmonised quality criteria should be facilitated.

In detail: Porthos[159] — a Project of Common Interest

Porthos, a joint venture between the Port of Rotterdam Authority, Gasunie and EBN, is aiming to transport CO2 and store it in empty gas fields beneath the Nord Sea. The CO2 will be captured by companies in the industrial cluster of the Port of Rotterdam and will be supplied to a collective pipeline. The collected gas will be then pressurised and transported through an offshore pipeline to a platform in the North Sea. The system will be operational by 2024 when it is expected to be able to store 2.5 million tons of CO2 per year. Furthermore, it involves a cross border element as it includes CCS partnership between Rotterdam, Antwerp and North Sea Port.

The project has been recognised as a Project of Common Interest and receives funding from the Connecting Europe Facility. The European Commission has proposed funding of EUR 102 million for this purpose.

Northern Lights[160] — a Project of Common Interest

Northern Lights is the combines CO2 transport and storage part of Norway's full-scale CCS value chain project by 2024. As part of the Northern Lights project, liquefied and pressurised CO2 will be captured and shipped from the capture sites to the onshore intermediate storage, which will be followed by the transport of CO2 through pipeline to the subsea well for injection into a geological storage complex.

The project will have the capacity to receive further volume of CO2 from Europe, beyond the 800 000 tons received from the Norwegian CCS Full-Scale Project. The Northern Light project was recognised as Project of Common Interest in January 2020.

Conclusions: Case study on Carbon Capture Utilisation and Storage

Commercialisation barriers majorly hinder CCUS deployment, while technology transfer is not considered as main concern, since the technology is already mature enough for deployment and the further developments are driven by the industry. However, deployment is slow due to the high uncertainty in terms of financial risk, lack of political commitment and high costs of first movers.

The EU could step in and provide sound policy framework, which could balance out the cyclical nature of policy changes. Moreover, it plays a crucial role in proving to private investors, that there is a business case with limited risk that can be factored in and priced accordingly by private capital.

Therefore, public investment should be better coordinated at regional, national and EU level supporting not only demonstration projects but also initial operating costs while providing guarantees against risks along the value chain and the time horizon. From policy makers, this would also require ambitious long-term policy framework with clear political commitment to boost public acceptance and fine-tuned regulations, which enable transport and storage of CO_2 with clear quality standards and defined liability criteria.

6. CONCLUSIONS AND RECOMMENDATIONS

While each case study has demonstrated the particularities of the respective technology, nonetheless some common trends emerge when comparing facilitators and barriers across these green technologies.

The technology transfer and commercialisation of green technologies requires substantial and long-term public support, ranging on a broad spectrum from the de-risking of investments, subsidies and guarantees addressing market failures to demonstrator activities. In the following, findings and recommendations are summarised along three dimensions: (1) the public policy framework; (2) the financial environment; and (3) the technology transfer process.

6.1 Public policy framework

Green technologies with the potential to bring about the systemic change require an enabling and stable regulatory framework to transform our societies and transition towards a zero-emission economy. To support the technology transfer and commercialisation of new green technologies, public policy primarily needs to provide predictable environment to businesses and investors thereby decreasing their investment risk.

In this regard, policy makers need to maintain a strong narrative and clear objectives to industry to support and incentivise the take-up of new green technologies. This requires clear public commitment to climate neutrality targets to create reliable business case for green technologies. In addition, policy makers should strive to foster the acceptance of new green technologies.

Without interfering unduly with the principle of technological neutrality, public policy needs to make clear choices when it comes to deploying novel green technologies. This means paving a clear path for high-impact technologies to be scaled-up and deployed within a fixed timeframe. Many stakeholders point firstly at the need to quickly deploy mature, green technologies to ensure economies of scale and bring down the costs of the specific technology. In this context, public policy needs to address potential market failures and support the scale-up of high potential green technologies in view of reaching ambitious climate targets. In this context, the public sector could consider to support key technologies through public procurement and investments in relevant infrastructure.

The scaling-up of mature green technologies is needed to lay the groundwork for the further technology transfer and commercialisation of green technologies, in particular via a market-pull effect of next generation technologies. This requires more than traditional research policy tools. Especially, in the case of green technologies requiring systemic change in their respective ecosystem to succeed, public policy support needs to be specifically targeted towards the needs of the respective technology and its industrial ecosystem. In addition, the policy support needs to address adequately all stages of the technology transfer and commercialisation process.

In many areas, unclear or outdated regulation remain a barrier to new and emerging technologies, especially if existing regulations are not able to provide the required regulatory certainty. Moreover, gaps in the regulatory framework for novel technologies need to be identified and filled. Flexibility and reactivity are required to adapt regulation to reflect the needs of the specific green technology.

The policy approach should follow an integrated approach along the entire value chain of the specific green technology, reflecting the needs of green technologies with a systemic reach. Research and innovation policy, industrial policy and environmental policy and their respective tools and objectives should not act in silos but be joined up into an integrated public policy approach. In particular, research and innovation policy needs to be interlinked with industrial policy to ensure the success of green technologies with a systemic reach.

The commitment of the European Commission under the new ERA Strategy to support the implementation of the EU Industrial Strategy by developing common industrial technology roadmaps by the end of 2022, linking key partnerships under Horizon Europe with industrial ecosystems, seems to be a major step forward in this regard.[161]

There is a high EU added value in the fostering of the key enabling technologies of the green transition. On the one hand, the role of the EU institutions is key to avoid regulatory fragmentation across the Union and to provide a common regulatory environment across the single market. On the other hand, EU-wide sectoral initiatives, such as the European Battery Alliance or the Fuel Cells and Hydrogen Joint Undertaking, demonstrate the high benefit of joint EU-level initiatives bringing together industry and research while providing a direct link to EU policy, regulation and funding. In bringing together industry, researchers, policy makers and investors, EU initiatives can help to kick-start green technologies at EU-level and to develop joint deployment strategies.

6.2 Financial environment

Investment in green technologies must be profitable. If such investment cannot generate profit, there will be no business case, and the public support will be insufficient. In general, the initial investment costs of green technology appear to be higher than those of conventional technologies and the time horizon of bringing green technologies to the market is a long term one, potentially from 10 to 20 years. Thus, investors need a clear sign of political commitment over the political cycle, that creates certainty regarding the long-term future of the technology and reduces the risk of investment.

Public support is needed and should focus on creating the business case for green technologies in order to crowd in private capital. Investors need to see where the revenue streams will come from, how will the demand for the technology evolve, and how the dependence on public sources, enabling infrastructure and third parties will affect the chances of their investment's profitability. Therefore, public support should focus on the realisation of demonstration projects and go even further and support enabling infrastructure across the value chain to reduce third-party risk and enable market uptake. It should also cover operating costs in the first years in order to make sure that first movers do not suffer disproportionate losses. Blended financing is needed, which should be phased out gradually while crowding-in private capital and providing public guarantees against long-term and third-party risk. The use of different funds at the regional, national or EU levels should be better coordinated in a way that public support should accompany these projects along the investment horizon: starting with early-stage financing, through demonstration projects until only guarantees for the viability of deployment are needed.

Contrary to what the literature suggested, financial institutions and innovators do not struggle to get connected due to the fragmentation or the absence of a marketspace. However, finding mature start-ups with mature products is indeed a barrier. Therefore, more long-term public financing schemes could be provided for start-ups in order to reach higher maturity level. This should include early-stage financing – as access to finance for low TRL technologies still remains an issue - combined with a push for increased financial transparency and literacy. This could also prevent the premature sell of start-ups with high potential.

Financial institutions and investors would need to assess and put a price tag on the risks, but they do not have the necessary data and tools available. As for new technologies, it is common that there is no robust historical data, but green technologies are also hard to be compared with any other technological investment. Therefore, richer datasets are needed, which could be addressed through clear reporting requirements and guidelines for new risk models factoring in non-financial risks could help the assessment of green technologies.

Lastly, in order to achieve the ambition of the European Green Deal, policy makers need to consider tools to facilitate the uptake of state-of-art green technologies. In some cases, industry is deploying technologies that have been available for many years and have less interest in experimenting with the application of novel technologies, therefore, they do not really consider technology transfer from RTOs as an integral part of their activities. Stepping up public support for these state-of-art green technologies would imply higher risk for the public, which is understandably not desirable. Nevertheless, if policy makers are ready to raise the level of ambitions, then more support for state-of-art innovation would be needed, which might require the public to shoulder increased risk for a limited period of time.

6.3 Technology transfer process

While research plays a key role in creating novel green technologies, these technologies will not achieve their potential of addressing climate change, reducing pollution or supporting the transition to circular economy, if they only stay in the labs or on the pages of scientific journals. Therefore, the design of public research policies and internal RTOs' policies should actively support the implementation of research results in practice.

Technology transfer offices (TTOs) play an indispensable role in steering the technology transfer process with their knowledge, experience and contacts, thereby propelling research results towards their market uptake. To enable the proper functioning of TTOs, it is thus crucial that they have sufficient resources. In this regard, the financing of TTOs should be systematic instead of only project based. This can enable TTOs to focus on projects with the highest potential instead of short-term revenue generation and partially disincentivise IP overvaluation resulting from the need to create revenue.

In general, as the success of technology transfer and commercialisation is often people-centric, entrepreneurial mind set, good team and management continuity throughout the commercialisation project are key to successfully commercialising a (green) technology.

The report showed several illustrations of how important informal and semi-formal stakeholder networks are. It showed in particular the value of open innovation, collaboration between researchers and the industry, which often results in joint research activities, research targeted to the market needs, knowledge spill-overs, and more effective technology transfer. Therefore, researchers, as well as staff of technology transfer offices should actively engage in networking activities with other stakeholders as a basis for the potential future collaboration. In this regard, strong public support to foster new cross-European stakeholder networks could play an important role.

When it comes to the technology transfer of new and potentially ground-breaking technologies, universities most often find their partners in start-ups and spin-offs. Even though these entities may have greater difficulties in leveraging sufficient finance to commercialise new-to-market innovation, they are typically more willing to shoulder the risks linked to the uptake of unproved new-to-market technological solutions. On the other hand, larger businesses frequently cooperate with universities on the optimisations of their technologies. When it comes to SMEs, the successful cooperation with RTOs often flounders on the lack of interest, typically caused by SME's risk-averse attitude, limited IP knowledge[162] and lack of experience with technology transfer. To successfully partner with universities in the commercialisation process, this common knowledge gap of SMEs needs to be closed.

To successfully transfer and commercialise green technologies, the licensing or sale of a patent is often insufficient. Technical knowhow and trade secrets play an indispensable role in the technology transfer as patents tend to have a narrower scope and additional knowledge is thus typically needed to replicate the invention.

The field of green technologies is characterised by a healthy patent competition in most areas. As there typically exist several solutions on how to solve a certain technical problem, a patent on one of them does not block the commercialisation of other substitutes. Therefore, IP-related issues cannot be considered a significant barrier for most green technologies (compared to other technologies).

Green technologies are often complex inventions that need to prove their commercial viability through demonstration and testing before being mass-produced and brought to the market. To this end, the existence of the right infrastructure - pilot and demonstration facilities and production lines - is central to prove the technology concept, develop a prototype and prove suitability of the technology for mass production.

The construction of open access demonstration facilities for green technologies is considered an essential public investment since start-ups and SMEs often lack capacities for in-house testing and prototyping green technologies.

Furthermore, the diffusion of green technologies cannot stop at the point where mature technology is available. Technologies need to be often further developed to be integrated within existing systems (such as electrical grid) or otherwise evolve to meet market expectation and appeal to consumers.

The commercialisation and diffusion of technologies can be a very long process, especially for complex and investment-heavy green technologies at the early stages of their development. Financial support for the commercialisation of green inventions should thus be available spanning the whole maturing process of the technology. Indeed, much of the green innovation needed to achieve climate neutrality is still at prototype or demonstration phase,[163] there is thus an imminent need to support the deployment and diffusion of green technologies to hit the ambitions environmental targets. Moreover, also the diffusion time of green technologies, notably those currently at an early stage of their development, has to be significantly reduced compared with historical averages. Therefore, the strong involvement of all actors – RTOs, TTOs, industry, private investors, public funding, the EU and Member States as well as prosumers and consumers – in technology commercialisation, diffusion and mainstreaming will be key to achieving the central Green Deal objective of reaching climate neutrality by 2050.

6.4 Recommendations to facilitate technology transfer and commercialisation

Based on the four case studies conducted for this report, the following recommendations emerged regarding the support of technology transfer and commercialisation of green technologies.

Public Policy Framework

Key recommendations

01

Public policy should provide an enabling and stable regulatory framework providing long-term regulatory certainty. EU regulation and coordination with Member States is needed to avoid regulatory fragmentation and ensure technical interoperability across borders.

02

The policy approach should follow a holistic, integrated approach along the entire value chain, reflecting the needs of green technologies with a systemic reach. Public policy needs to provide a clear path for high-impact green technologies to be scaled-up and deployed within a fixed timeframe. Research & innovation policy, industrial policy and environmental policy and their respective tools and objectives should not act in "silos" but be joined up into an integrated public policy approach.

03

EU institutions should continue to play an active role to realise the EU added value for green technologies by strengthening networks bringing together industry, researchers, investors and policy makers. Such action should also benefit niche research areas, with a high potential to contribute to the European Green Deal objectives.

04

Policy makers should provide a strong narrative and clear objectives (*e.g.* climate neutrality targets) and long-term commitment to industry to support and incentivise the take-up of new green technologies and to increase technology acceptance. In relevant technology areas, the public sector could consider venues to support viable green technologies through infrastructures and public procurement.

05

Policy makers should prove a high degree of reactivity in addressing regulatory gaps and unclear or outdated regulation, which remain a barrier to new and emerging technologies.

Financial Framework

Key recommendations

01

Public support should bring down first movers' costs for green technologies. In this regard, public support should bridge gaps in private funding for high-impact technologies at low technology readiness levels. Financial support should cover demonstration projects and commercialisation (including CAPEX, OPEX and enabling infrastructure) through blended financing and guarantee instruments.

02

Financing at the regional, national and European level (including the EU's funding instruments) should be coordinated more closely and should better address long-term and third-party risks.

03

The availability of dedicated green funds, loan and guarantee schemes should be increased and public-private partnerships with a high public visibility should be promoted to show a high level of political commitment and foster technology acceptance in investors and the wider public. Where appropriate, policy makers should consider standards and product-labels to generate public trust.

04

The EU should develop data reporting guidelines for green technologies and new risk assessment models that factor in non-financial risk, such as the effects of climate change. Financial institutions need to be given better toolkits to assess risks related to investments in green technologies.

05

Policy makers should take steps to develop strategic frameworks for increasing financial planning literacy of start-ups and SMEs regarding the commercialisation of innovative green technologies combined with raising awareness of available financial instruments.

Technology transfer process

Key recommendations

01

Policies should factor in the needs and characteristics of the respective technology and its industrial ecosystem. This means addressing all stages of the technology transfer and commercialisation process, from the early research stages up to mature technologies.

02

The availability and use of real-world testing environments and regulatory sandboxes should be increased to permit innovation to reach market readiness and ensure higher success rates when deployed to the market.

03

Technology transfer and commercialisation activities should form a prominent part of the design of public research policies, as well as a strategic priority for RTOs. To that effect, sufficient funding should be ensured both for the technology transfer process and for technology transfer offices.

04

Through training and awareness-raising activities, public institutions should promote greater awareness among stakeholders, and in particular SMEs, about IP rights in the commercialisation process, their protection, valuation and licensing, with a focus on the specificities of green patents.

05

Open innovation models should be studied with a focus on the needs of various green technologies. In particular, it should be analysed whether pooling green patents would be desirable and could support the deployment of green innovation in certain sectors. Such an analysis should build on the existing experience with creating green innovation marketplaces and patent pools.

GLOSSARY

AGI: Artificial general intelligence

AI: Artificial intelligence

BDVA: Big Data Value Association

CEF: Connecting Europe Facility

CCS: Carbon capture and storage

CCU: Carbon capture and utilisation

CCUS: Carbon capture and storage and utilisation

CLAIRE: Confederation of Laboratories for Artificial Intelligence Research in Europe

CO2: Carbon dioxide

EIB: European Investment Bank

EIF: European Investment Fund

EIT: European Institute of Innovation and Technology

ELISE: European Network of AI Excellence Centres

EPO: European Patent Office

ERA: European Research Area

ETIP: European Technology and Innovation Platform

ETS: Emissions Trading System

EU: European Union

EuRobotics: European Robotics Association

EurAi: European Association for Artificial Intelligence

EV: Electric vehicle

GDPR: General Data Protection Regulation

GHG: Greenhouse gases

IEA: International Energy Agency

IP: Intellectual property

IPC: International Patent Classification

IPCEI: Important Projects of Common European Interest

IPO: Initial public offering

JRC: Joint Research Centre

Li-ion battery: Lithium-ion battery

MBS: Mortgage-backed securities

MFF: Multiannual Financial Framework

ML: Machine learning

NASA: National Aeronautics and Space Administration

NDA: Non-disclosure agreement

OECD: Organisation for Economic Co-operation and Development

R&D&I: Research, development, and innovation

R&I: Research and innovation

RTOs: Research and technology organisations

SEP: Standard Essential Patent

SME: Small and medium-sized enterprise

TEN-E: Trans-European Networks for Energy

TRL: Technology readiness levels

TTO: Technology transfer offices

USPTO: United States Patent and Trademark Office

WIPO: World Intellectual Property Organization

ENDNOTES

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14 European Commission (2020). *A new ERA for Research and Innovation.* COM(2020) 628 final.

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16 See actions 6 and 7 in European Commission (2020). *A new ERA for Research and Innovation.* COM(2020) 628 final.

17 European Patent Office (2020). *Valorisation of scientific results. Patent commercialisation scoreboard: European universities and public research organisations.* Retrieved January 9, 2021, from: https://eua.eu/images/site1/team/EPO_universities_key-findings_en_1811_002.pdf.

18 See *e.g.* the European Patent Convention or the forthcoming EU unitary patent system, or internationally the Paris Convention for the Protection of Industrial Property, Patent Cooperation Treaty, and the Agreement on Trade-Related Aspects of Intellectual Property Rights (TRIPS).

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20 European Union Intellectual Property Office (2017). *Protecting Innovation Through Patents and Trade Secrets: Determinants and Performance Impacts for European Union Firms.* Retrieved January 9, 2021, from: https://euipo.europa.eu/tunnel-web/secure/webdav/guest/document_library/observatory/documents/reports/Trade%20Secrets%20Report_en.pdf.

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22 *Ibid.*

23 *Ibid.*

24 Hernández, H., Grassano, N., Tübke, A., Amoroso, S., Csefalvay, Z., Gkotsis, P. (2020). *The 2019 EU Industrial R&D Investment Scoreboard.* Publications Office of the European Union, Luxembourg.

25 Calculation based on the patents filed at the EPO and USPTO over the period 2012-2015. *Ibid.: The 2019 EU Industrial R&D Investment Scoreboard.*

26 United Nations Environment Programme, European Patent Office (2015). *Climate change mitigation technologies in Europe – evidence from patent and economic data.* Retrieved January 9, 2021, from: http://documents.epo.org/projects/babylon/eponet.nsf/0/6A51029C350D3C8EC1257F110056B93F/$File/climate_change_mitigation_technologies_europe_en.pdf

27 European Commission (2020). *Commission Staff Working Document. Clean Energy Transition – Technologies and Innovations. Accompanying the document Report from the Commission to the European Parliament and the Council on progress of clean energy competitiveness.* COM(2020) 953 final.

28 While no formal definition of "green technology patents" exists, the IPC Committee of Experts established under the 1971 Strasbourg Agreement Concerning the International Patent Classification developed a non-exhaustive "IPC Green Inventory" containing patents relating to environmentally sound technologies, as listed by the UNFCCC. EPO has established its own classification of "green patents" – Y02. Even though each classification has its specificities, for the reasons of simplicity, they will be used interchangeably in this study.

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31 European Commission (2020). *Factsheet: Valorisation. Making results work for society.* Retrieved January 9, 2021, from: https://ec.europa.eu/info/sites/info/files/research_and_innovation/strategy_on_research_and_innovation/documents/ec_rtd_valorisation_factsheet.pdf.

32 *Ibid.* United Nations Environment Programme, European Patent Office. Climate change mitigation technologies in Europe – evidence from patent and economic data.

33 *Ibid.*

34 OECD (2015). *Technology diffusion data. Retrieved January 9, 2021, from: https://www.oecd.org/env/indicators-modelling-outlooks/green-patents.htm.* Slightly better convergence rate sees Germany, the Europe´s leader in green patenting, holding a 12.6% share of developed and 9.9% share of patented world´s green innovation.

35 This has been also acknowledged in the recent ERA Strategy that announced that the Commission will "develop and test a networking framework in support of Europe's R&I ecosystems to strengthen excellence and maximise the value of knowledge creation, circulation and use". For details see European Commission (2020). *A new ERA for Research and Innovation.* COM(2020) 628 final.

36 This cooperation can be documented by the rise in co-patented inventions. In 2014, the European Patent Office counted 948 co-patent applications by universities and industry, *i.e.* 43% of all European Patent Office patent applications. The share of co-patented inventions has nearly doubled since 1992. In: European Commission (2020). ***Factsheet: Valorisation.*** Making results work for society. (*Ibid.).*

37 https://ip-marketplace.org/.

38 For more information see: https://www3.wipo.int/wipogreen/en/.

39 https://medicinespatentpool.org/.

40 For details as to some of the challenges faced by GreenXchange and details about its mode of operation see Ghafele R., O´Brien R.D. (2012). *Open innovation for sustainability: Lessons from the GreenXchange experience. International Journal of Economic Behavior and Organization.* Vol.1 (Issue 1).

41 Taubman A. (2009). *Sharing Technology to Meet a Common Challenge. WIPO Magazine.* Retrieved on 9 January 2021 from: https://www.wipo.int/wipo_magazine/en/2009/02/article_0002.html

42 The interviewed stakeholders shared a careful optimism that such green patent marketplace could be useful, however, only if it is well-structured and well-managed, includes all necessary information to make it user-friendly and if it gathers enough green patents to be relevant.

43 The European Digital Innovation Hubs for the diffusion of digital technologies could be one possible inspiration for activities to be piloted in the area of green technologies. Other already existing platforms that bring together businesses and research centers and offer acceleration and consulting services are EIT Innovation Communities.

44 This would however limit the spill-over effect among sectors.

45 These can be sectoral standards for a given green technology or technology neutral. Such as a standard for energy efficiency of buildings. The latter can create considerable market pull and business case for the deployment of green technologies.

46 Dinnetz, M. (2018): Technology Transfer: From research to Impact, European Commission, Luxembourg.

47 Kraemer-Eis, H.. et Al (2009).: EIF Research and Market Analysis: Financing technology transfer, Luxembourg.

48 Ferreri, Annarita; Grande, Sergio (2016): Approaches to and methods for evaluating new technologies in Technology Transfer Offices: How long is a piece of string?, JRC Conference and Workshop Reports, Luxembourg.

49 Kaldos, Peter (2011): IP valuation at research institutes: An essential tool for technology transfer, HIPO, Budapest.

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51 International Energy Agency. (2020). *Clean Energy Innovation. Innovation Needs in the Sustainable Development Scenario.* Retrieved January 9, 2021, from: https://www.iea.org/reports/clean-energy-innovation/innovation-needs-in-the-sustainable-development-scenario#reference-3.

52 According to Nemet et al., a demonstration facility, especially in energy and heavy industrial sectors, may require investments of USD 1 billion or more. See Nemet, G., Kraus, M. and Zipperer, V. (2016). *The Valley of Death, the Technology Pork Barrel, and Public Support for Large Demonstration Projects.* Retrieved January 9, 2021, from: https://econpapers.repec.org/paper/diwdiwwpp/dp1601.htm.

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54 Hydrogen Council (2017). ***Hydrogen scaling up. A sustainable pathway for the global energy transition.*** Retrieved January 10, 2021, from https://hydrogencouncil.com/wp-content/uploads/2017/11/Hydrogen-Scaling-up_Hydrogen-Council_2017.compressed.pdf.

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56 Hydrogen Europe (2020). ***Clean Hydrogen Monitor 2020.*** Retrieved January 10, 2021, from https://www.hydrogeneurope.eu/node/1691.

57 Kanellopoulos, K., Blanco Reano, H. (2019). ***The potential role of H2 production in a sustainable future power system - An analysis with METIS of a decarbonised system powered by renewables in 2050.*** Publications Office of the European Union, Luxembourg.

58 Hydrogen Europe (2020). ***Clean Hydrogen Monitor 2020.*** Retrieved January 10, 2021, from https://www.hydrogeneurope.eu/node/1691.

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60 European Commission (2019). ***Strengthening Strategic Value Chains for a future-ready EU Industry. Report of the Strategic Forum for Important Projects of Common European Interest Strategic Forum.*** Retrieved January 10, 2021, from https://ec.europa.eu/docsroom/documents/37824.

61 European Commission (2020). ***A hydrogen strategy for a climate-neutral Europe.*** COM(2020) 301 final.

62 See for example national strategies on hydrogen in France, Germany and the Netherlands.

63 European Commission (2019). ***Inception impact assessment. Ares(2019)4972390.*** Retrieved January 10, 2021, from https://ec.europa.eu/info/law/better-regulation/have-your-say/initiatives/11902-Energy-European-Partnership-for-clean-hydrogen-Horizon-Europe-programme-.

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79 *Ibid.*

80 International Energy Agency (2019). *The Future of Hydrogen. Seizing today's opportunities. IEA, Paris.* Retrieved January 10, 2021, from https://www.iea.org/reports/the-future-of-hydrogen, p. 13.

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82 The European market for batteries is projected to grow to EUR 250 billion per year by 2025. See https://www.eba250.com/.

83 According to some calculations, the total battery supply-chain investments from European governments, manufacturers, development banks and commercial lenders could top EUR 100 billion in 2020.

84 The European share of global cell manufacturing is projected to increase from 3% in 2018 to more than 15% by 2028. See: European Commission (2019). *Report from the Commission on the Implementation of the Strategic Action Plan on Batteries: Building a Strategic Battery Value Chain in Europe.* COM(2019) 176 final.

85 As the joint EPO and IEA report states, Japan and the Republic of Korea are leading the global battery technology race, pushing other countries to develop competitive advantages in specific parts of the battery value chain. Of the top ten global applicants behind international patent families related to batteries, nine are based in Asia. German company Bosch is the only European representative. Generally, more than half of patent families in electricity storage originating from Europe are from Germany. See European Patent Office, International Energy Agency (2020). *Innovation in batteries and electricity storage - A global analysis based on patent data.* Retrieved January 9, 2021, from: http://documents.epo.org/projects/babylon/eponet.nsf/0/969395F58EB07213C12585E7002C7046/$FILE/battery_study_en.pdf

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92 It will provide an opportunity to produce, test and demonstrate innovative battery technologies at scale, helping to bridge the gap between research and innovation results (for instance achieved in Horizon 2020), and commercial deployment of battery manufacturing.

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99 In 2018, innovation in Li-ion cells was responsible for 45% of patenting activity related to battery cells, compared with just 7.3% for cells based on other chemistries. See: European Patent Office, International Energy Agency (2020). Innovation in batteries and electricity storage - A global analysis based on patent data. (*Ibid.).*

100 Lithium-ion batteries were invented 30 years ago but the current boom it experiences outside of the consumer electronics has been concentrated in the last couple of years. It is thus expected that even with the increased pace of innovation, any new battery beyond Li-ion chemistry would take at least 10 years to get to the market.

101 See *e.g.* the Lipping University recently published its discovery of a new "organic" redox-flow battery. Or the JRC report on the future battery technologies: Lebedeva N., Ruiz Ruiz V., Bielewski M., Blagoeva D., Pilenga A., (forthcoming). Selected Advanced Lithium-ion, Post Lithium-ion and Non-Lithium-ion Batteries Technology Development Report.

102 Patenting activity related to battery recycling remained at relatively low levels with a total of 436 international patent families between 2000 and 2018. See: European Patent Office, International Energy Agency (2020). Innovation in batteries and electricity storage - A global analysis based on patent data. (*Ibid.).*

103 The price of Li-ion batteries drops on average by 20% a year – be it in part due to new chemistries, mostly adjustments to the composition of the battery cathode, and in part due to the economies of scale in manufacturing. Innovation in manufacturing processes also plays a role. According to the joint EPO and IEA report, patenting activity in the manufacturing of battery cells and cell-related engineering developments has grown threefold over the last decade. See: European Patent Office, International Energy Agency (2020). Innovation in batteries and electricity storage - A global analysis based on patent data. (*Ibid.).*

104 According to the joint EPO and IEA report, innovation in battery technology is still mainly concentrated within a limited group of large companies, which generated a stable share of about 80% of major innovations related to batteries between 2000 and 2018. The remaining share is split almost equally between SMEs and universities and public research organisations. See: European Patent Office, International Energy Agency (2020). Innovation in batteries and electricity storage - A global analysis based on patent data. (*Ibid.).*

105 A prototype Li-ion battery was developed in 1985, based on earlier university research during the 1970s–1980s. The inventors behind Li-ion battery were awarded by the Nobel Prize for this invention in 2019.

106 As compared to SMEs that account for 15.9% of the patent families. See: European Patent Office, International Energy Agency (2020). Innovation in batteries and electricity storage - A global analysis based on patent data. (*Ibid.*).

107 https://battery2030.eu/.

108 The smart functionalities include *e.g.* sensors probing directly at the battery cell level possible degradation or hazardous reactions and triggering the release of self-healing agents that can control such unwanted reactions.

109 As one of the interviewees summarised: "At some point, it is necessary to shift the focus from researching ever-better batteries to the actual implementation of the solutions which we already have.".

110 For more information see: https://www.he3da.com/copy-of-domu.

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